
Physics 523 Final

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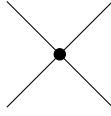
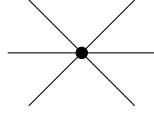
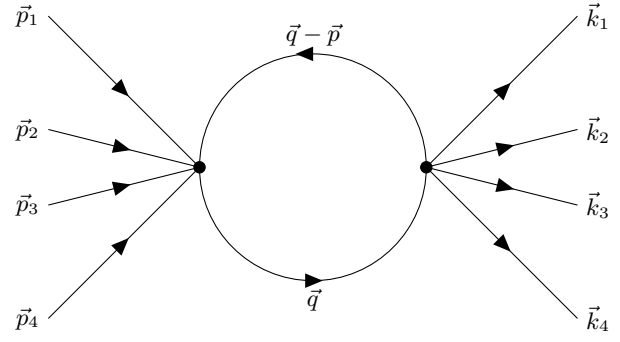
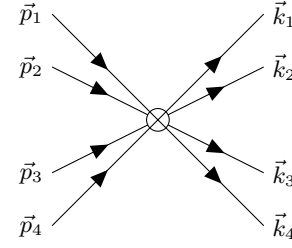

 FIG. 1. Coefficient: $-6i\lambda$

 FIG. 2. Coefficient: $-120ig$

 FIG. 3. $\star$


FIG. 4. This counterterm vertex does not exist.

Part I

Problem 1

1. LAGRANGIAN

$$\mathcal{L} = \frac{(\partial\phi)^2}{2} - \frac{m^2}{2}\phi^2 - \frac{\lambda}{4}\phi^4 - \frac{g}{6}\phi^6 \quad (1)$$

2. UNITS

$$[\mathcal{L}] = n, \quad [\phi] = \frac{n}{2} - 1 \quad (2)$$

$$[\mathcal{L}] = [\lambda\phi^4] \implies [\lambda] = n - 4[\phi] = 4 - n \quad (3)$$

$$[\mathcal{L}] = [g\phi^6] \implies [g] = n - 6[\phi] = 6 - 2n \quad (4)$$

$$\boxed{[\lambda] = 4 - n}, \quad \boxed{[g] = 6 - 2n} \quad (5)$$

In 4D $[\lambda] = 0$ and $[g] = -2$.

3. FEYNMAN RULES

The propagator is

$$\frac{i}{p^2 - m^2 + i\epsilon}. \quad (6)$$

The interactions are in Figs. 1 & 2.

4. RENORMALIZATION

This Lagrangian does not have an 8-point interaction (ϕ^8). If the one-loop 8-point graph (Fig. 3) diverges there is no counterterm (Fig. 4) available to swallow the infinity.

$$\vec{p} = \sum_{n=1}^4 \vec{p}_n = \sum_{n=1}^4 \vec{k}_n \quad (7)$$

5. LOOP DIAGRAM

$$\star = \frac{(-120ig)^2}{(2\pi)^4} S \int d^4q \frac{i}{q^2 - m^2 + i\epsilon} \frac{i}{(\vec{q} - \vec{p})^2 - m^2 + i\epsilon} \quad (8)$$

S is the symmetry factor.

$$\star = \frac{120^2}{(2\pi)^4} g^2 S \int \frac{d^4q}{(q^2 - m^2 + i\epsilon)[(\vec{q} - \vec{p})^2 - m^2 + i\epsilon]} \quad (9)$$

$$\star \propto \int \frac{d^4q}{(q^2 - m^2 + i\epsilon)[(\vec{q} - \vec{p})^2 - m^2 + i\epsilon]} \quad (10)$$

5.1. Feynman Parameters

$$A = (\vec{q} - \vec{p})^2 - m^2 + i\epsilon, \quad B = q^2 - m^2 + i\epsilon \quad (11)$$

$$\star \propto \int \frac{d^4q}{AB} = \int d^4q \int_0^1 \frac{dx}{D^2} = \int_0^1 dx \int \frac{d^4q}{D^2} \quad (12)$$

$$\frac{1}{AB} = \int_0^1 \frac{dx}{D^2}, \quad D = A + (B - A)x \quad (13)$$

$$D = (\vec{q} - \vec{p})^2 + [q^2 - (\vec{q} - \vec{p})^2]x - m^2 + i\epsilon \quad (14)$$

$$D = (\vec{q} - \vec{p})^2 + 2x\vec{q} \cdot \vec{p} - xp^2 - m^2 + i\epsilon \quad (15)$$

$$D = q^2 - 2(1-x)\vec{q} \cdot \vec{p} + (1-x)p^2 - m^2 + i\epsilon \quad (16)$$

5.2. Complete the Square

$$ax^2 + bx + c = a \left(x + \frac{b}{2a} \right)^2 + c - \frac{b^2}{4a} \quad (17)$$

$$D = [\vec{q} - (1-x)\vec{p}]^2 - \Delta + i\epsilon \quad (18)$$

$$\Delta = m^2 - (1-x)xp^2 \quad (19)$$

5.3. Shift Integrand

$$\star \propto \int_0^1 dx \int \frac{d^4 q}{([\vec{q} - (1-x)\vec{p}]^2 - \Delta + i\epsilon)^2} \quad (20)$$

$$\vec{k} = \vec{q} - (1-x)\vec{p}, \quad d^4 q = d^4 k \quad (21)$$

$$\star \propto \int_0^1 dx \int \frac{d^4 k}{(k^2 - \Delta + i\epsilon)^2} \quad (22)$$

5.4. Divergence

$$\star \sim \int_1^\Lambda dk \frac{k^3}{k^4} = \int_1^\Lambda \frac{dk}{k} = \ln \Lambda, \quad \Lambda \rightarrow \infty \quad (23)$$

6. CONCLUSION

Diagram $\star$ diverges so this theory is non-renormalizable.
QED.

Part II

Problem 2

7. LAGRANGIANS

The bare Lagrangian is

$$\mathcal{L} = \bar{\psi}_B(i\not{\partial} + m_B)\psi_B + \frac{(\partial\phi_B)^2}{2} - \frac{\mu_B^2}{2}\phi_B^2 - g_B\bar{\psi}_B\psi_B\phi_B. \quad (24)$$

$$(\partial\phi)^2 = \eta^{\alpha\beta}(\partial_\alpha\phi)(\partial_\beta\phi) \quad (25)$$

$$\begin{aligned} \psi_B &= \sqrt{Z_2}\psi, & m_B &= \sqrt{Z_m}m, & \phi_B &= \sqrt{Z_3}\phi, \\ \mu_B &= \sqrt{Z_\mu}\mu, & g_B &= Z_1g \end{aligned} \quad (26)$$

$$\begin{aligned} \mathcal{L} &= Z_2\bar{\psi}(i\not{\partial} + \sqrt{Z_m}m)\psi + Z_3\frac{(\partial\phi)^2}{2} - Z_3Z_\mu\frac{\mu^2}{2}\phi^2 \\ &\quad - Z_1Z_2\sqrt{Z_3}g\bar{\psi}\psi\phi \end{aligned} \quad (27)$$

$$\begin{aligned} \mathcal{L} &= Z_2i\bar{\psi}\not{\partial}\psi + Z_2\sqrt{Z_m}m\bar{\psi}\psi + Z_3\frac{(\partial\phi)^2}{2} - Z_3Z_\mu\frac{\mu^2}{2}\phi^2 \\ &\quad - Z_1Z_2\sqrt{Z_3}g\bar{\psi}\psi\phi \end{aligned} \quad (28)$$

$$Z_4 = Z_2\sqrt{Z_m}, \quad Z_5 = Z_1Z_2\sqrt{Z_3} \quad (29)$$

$$\mathcal{L} = Z_2i\bar{\psi}\not{\partial}\psi + Z_4m\bar{\psi}\psi + Z_3\frac{(\partial\phi)^2}{2} - Z_3Z_\mu\frac{\mu^2}{2}\phi^2 - Z_5g\bar{\psi}\psi\phi \quad (30)$$

$$Z_i = 1 + \delta_i \quad (31)$$

$$\begin{aligned} \mathcal{L} &= (1 + \delta_2)i\bar{\psi}\not{\partial}\psi + (1 + \delta_4)m\bar{\psi}\psi + (1 + \delta_3)\frac{(\partial\phi)^2}{2} \\ &\quad - (1 + \delta_3)(1 + \delta_\mu)\frac{\mu^2}{2}\phi^2 - (1 + \delta_5)g\bar{\psi}\psi\phi \end{aligned} \quad (32)$$

$$\begin{aligned} \mathcal{L} &= (1 + \delta_2)i\bar{\psi}\not{\partial}\psi + (1 + \delta_4)m\bar{\psi}\psi + (1 + \delta_3)\frac{(\partial\phi)^2}{2} \\ &\quad - (1 + \delta_\mu + \delta_3 + \delta_3\delta_\mu)\frac{\mu^2}{2}\phi^2 - (1 + \delta_5)g\bar{\psi}\psi\phi \end{aligned} \quad (33)$$

I drop $\mathcal{O}(\delta^2)$ terms.

$$\begin{aligned} \mathcal{L} &= (1 + \delta_2)i\bar{\psi}\not{\partial}\psi + (1 + \delta_4)m\bar{\psi}\psi + (1 + \delta_3)\frac{(\partial\phi)^2}{2} \\ &\quad - (1 + \delta_\mu + \delta_3)\frac{\mu^2}{2}\phi^2 - (1 + \delta_5)g\bar{\psi}\psi\phi \end{aligned} \quad (34)$$

$$\begin{aligned} \mathcal{L} &= \bar{\psi}(i\not{\partial} + m)\psi + \frac{(\partial\phi)^2 - \mu^2\phi^2}{2} - g\bar{\psi}\psi\phi + \delta_2i\bar{\psi}\not{\partial}\psi \\ &\quad + \delta_4m\bar{\psi}\psi + \delta_3\frac{(\partial\phi)^2}{2} - \delta_\mu\frac{\mu^2}{2}\phi^2 - \delta_3\frac{\mu^2}{2}\phi^2 - \delta_5g\bar{\psi}\psi\phi \end{aligned} \quad (35)$$

The interaction Lagrangian is

$$\boxed{\mathcal{L}_1 = -g\bar{\psi}\psi\phi}. \quad (36)$$

$$\begin{aligned} \mathcal{L} &= \bar{\psi}(i\not{\partial} + m)\psi + \frac{(\partial\phi)^2 - \mu^2\phi^2}{2} + \mathcal{L}_1 + \delta_2i\bar{\psi}\not{\partial}\psi \\ &\quad + \delta_4m\bar{\psi}\psi + \delta_3\frac{(\partial\phi)^2}{2} - \delta_\mu\frac{\mu^2}{2}\phi^2 - \delta_3\frac{\mu^2}{2}\phi^2 - \delta_5g\bar{\psi}\psi\phi \end{aligned} \quad (37)$$

I integrate by parts.

$$(\partial\phi)^2 = \eta^{\alpha\beta}(\partial_\alpha\phi)(\partial_\beta\phi) = -\phi\eta^{\alpha\beta}\partial_\alpha\partial_\beta\phi \quad (38)$$

$$\eta^{\alpha\beta}\partial_\alpha\partial_\beta = \partial_t^2 - \nabla^2 = \square, \quad (\partial\phi)^2 = -\phi\square\phi \quad (39)$$

$$\begin{aligned} \mathcal{L} &= \bar{\psi}(i\not{\partial} + m)\psi - \frac{\phi(\square + \mu^2)\phi}{2} + \mathcal{L}_1 + \delta_2i\bar{\psi}\not{\partial}\psi \\ &\quad + \delta_4m\bar{\psi}\psi - \frac{\phi[\delta_3\square + (\delta_\mu + \delta_3)\mu^2]\phi}{2} - \delta_5g\bar{\psi}\psi\phi \end{aligned} \quad (40)$$

The free Lagrangian is

$$\boxed{2\mathcal{L}_0 = 2\bar{\psi}(i\not{\partial} + m)\psi - \phi(\square + \mu^2)\phi}. \quad (41)$$

The counterterm Lagrangian is

$$\begin{aligned} \mathcal{L}_C &= \bar{\psi}(i\delta_2\not{\partial} + \delta_4m)\psi - \frac{\phi[\delta_3\square + (\delta_\mu + \delta_3)\mu^2]\phi}{2} \\ &\quad - \delta_5g\bar{\psi}\psi\phi. \end{aligned} \quad (42)$$

$$\boxed{\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_1 + \mathcal{L}_C} \quad (43)$$

8. DIMENSIONS & DIMREG SCALING

8.1. 4D

$$[\phi] = 1, \quad [\psi] = \frac{3}{2}, \quad [\mathcal{L}] = 4 \quad (44)$$

$$\begin{aligned} [\mathcal{L}] &= [g\bar{\psi}\psi\phi] = [g\psi^2\phi] = [g] + 2[\psi] + [\phi] = [g] + 4 \implies \\ 4 &= [g] + 4 \implies [g] = 0 \end{aligned} \quad (45)$$

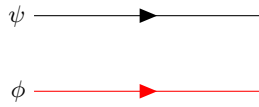


FIG. 5. Propagators

Particle	Spinor
incoming fermion	u_s
outgoing fermion	$\bar{u}_s$
incoming anti-fermion	$\bar{v}_s$
outgoing anti-fermion	v_s

TABLE I. Momentum eigenspinors for external fermions.

8.2. dD

$$[\phi] = \frac{d}{2} - 1, \quad [\psi] = \frac{d-1}{2}, \quad [\mathcal{L}] = d \quad (46)$$

$$[\mathcal{L}] = [g\bar{\psi}\psi\phi] \implies [g] = d - 2[\psi] - [\phi] = 2 - \frac{d}{2} \quad (47)$$

8.3. DimReg Scaling

To keep g dimensionless in d dimensions I introduce the scale

$$g \rightarrow g\beta^{(4-d)/2}, \quad [\beta] = 1, \quad (48)$$

so

$$[g\beta^{(4-d)/2}] = [g] + [\beta^{(4-d)/2}] = [\beta^{(4-d)/2}] = \frac{4-d}{2}[\beta] = 2 - \frac{d}{2}. \quad (49)$$

The Euler-Mascheroni constant is $\gamma \approx 0.577216$.

9. FEYNMAN RULES

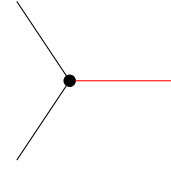
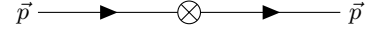
The propagators (Fig. 5) are

$$\phi \sim \frac{i}{p^2 - \mu^2 + i\epsilon}, \quad \psi \sim i \frac{\not{p} - m}{p^2 - m^2 + i\epsilon}. \quad (50)$$

External on-shell fermions are capped with momentum eigenspinors (Table I). The interaction vertex is in Fig. 6. The counterterm vertices are in Figs. 7-9.

9.1. Counterterm Coefficient

$$-\frac{\phi[\delta_3\Box + (\delta_\mu + \delta_3)\mu^2]\phi}{2} \rightarrow -\frac{2i[-\delta_3p^2 + (\delta_\mu + \delta_3)\mu^2]}{2} = i[\delta_3p^2 - (\delta_\mu + \delta_3)\mu^2] \quad (51)$$

FIG. 6. Coefficient: $-ig$ FIG. 7. Coefficient: $i(\delta_2\not{p} + \delta_4m)$, see Schwartz Eq. 19.13.

- 2 for 2 identical fields

- $\Box\phi \rightarrow -p^2$

10. φ SELF-ENERGY

I evaluate Fig. 10. The external particles are off-shell.

$$\heartsuit = -\frac{S}{(2\pi)^4} \int d^4q \quad \text{Tr} \left[-ig i \frac{\not{q} - \not{p} - m}{(\vec{q} - \vec{p})^2 - m^2 + i\epsilon} (-ig) i \frac{\not{q} - m}{q^2 - m^2 + i\epsilon} \right] \quad (52)$$

The symmetry factor is

$$S = \frac{2}{2!} = \frac{2}{2} = 1. \quad (53)$$

$$\heartsuit = -\frac{g^2}{(2\pi)^4} \int d^4q \frac{\text{Tr}[(\not{q} - \not{p} - m)(\not{q} - m)]}{[(\vec{q} - \vec{p})^2 - m^2 + i\epsilon](q^2 - m^2 + i\epsilon)} \quad (54)$$

$$\heartsuit = -\frac{g^2}{(2\pi)^4} \int d^4q \frac{N}{[(\vec{q} - \vec{p})^2 - m^2 + i\epsilon](q^2 - m^2 + i\epsilon)} \quad (55)$$

$$N = \text{Tr}[(\not{q} - \not{p} - m)(\not{q} - m)] \quad (56)$$

$$N = \text{Tr}(\not{q}^2 - m\not{q} - \not{p}\not{q} + m\not{p} - m\not{q} + m^2) \quad (57)$$

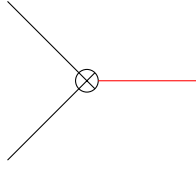
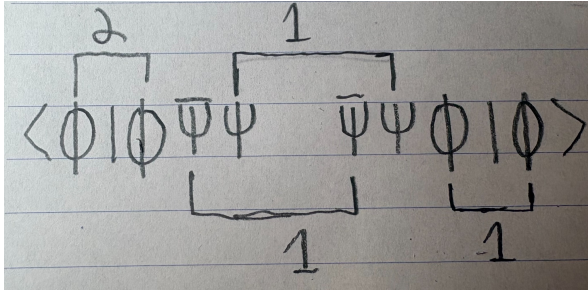
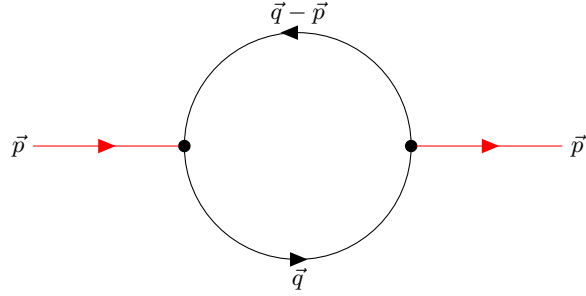
$$N = \text{Tr}(\not{q}^2 - 2m\not{q} - \not{p}\not{q} + m\not{p} + m^2) \quad (58)$$

$$N = \text{Tr} \not{q}^2 - \cancel{2m \text{Tr} \not{q}} - \text{Tr}(\not{p}\not{q}) + \cancel{m \text{Tr} \not{p}} + m^2 \text{Tr} I \quad (59)$$

I is the (usually implicit) identity matrix.

$$N = \text{Tr} \not{q}^2 - \text{Tr}(\not{p}\not{q}) + 4m^2 \quad (60)$$

FIG. 8. Coefficient: $i[\delta_3p^2 - (\delta_\mu + \delta_3)\mu^2]$, see Sec. 9.1.

FIG. 9. Coefficient: $-i\delta_5 g$ FIG. 10. Diagram $\heartsuit$ with symmetry factor combinatorics.

$$N = q_\alpha q_\beta \text{Tr}(\gamma^\alpha \gamma^\beta) - q_\beta p_\alpha \text{Tr}(\gamma^\alpha \gamma^\beta) + 4m^2 \quad (61)$$

$$\text{Tr}(\gamma^\alpha \gamma^\beta) = 4\eta^{\alpha\beta} \quad (62)$$

$$N = 4q_\alpha q_\beta \eta^{\alpha\beta} - 4q_\beta p_\alpha \eta^{\alpha\beta} + 4m^2 \quad (63)$$

$$N = 4(q^2 - \vec{q} \cdot \vec{p} + m^2) \quad (64)$$

$$\heartsuit = -\frac{4g^2}{(2\pi)^4} \int d^4q \frac{q^2 - \vec{q} \cdot \vec{p} + m^2}{[(\vec{q} - \vec{p})^2 - m^2 + i\epsilon](q^2 - m^2 + i\epsilon)} \quad (65)$$

$$\heartsuit = \frac{g^2}{4\pi^4} \int d^4q \frac{\vec{q} \cdot \vec{p} - q^2 - m^2}{[(\vec{q} - \vec{p})^2 - m^2 + i\epsilon](q^2 - m^2 + i\epsilon)} \quad (66)$$

10.1. Feynman Parameters

$$\heartsuit = \frac{g^2}{4\pi^4} \int d^4q \frac{\vec{q} \cdot \vec{p} - q^2 - m^2}{AB} \quad (67)$$

$$A = (\vec{q} - \vec{p})^2 - m^2 + i\epsilon, \quad B = q^2 - m^2 + i\epsilon \quad (68)$$

$$\frac{1}{AB} = \int_0^1 \frac{dx}{D^2}, \quad D = A + (B - A)x \quad (69)$$

$$\heartsuit = \frac{g^2}{4\pi^4} \int d^4q \int_0^1 dx \frac{\vec{q} \cdot \vec{p} - q^2 - m^2}{D^2} \quad (70)$$

$$\heartsuit = \frac{g^2}{4\pi^4} \int_0^1 dx \int d^4q \frac{\vec{q} \cdot \vec{p} - q^2 - m^2}{D^2} \quad (71)$$

$$D = (\vec{q} - \vec{p})^2 + [q^2 - (\vec{q} - \vec{p})^2]x - m^2 + i\epsilon \quad (72)$$

$$D = q^2 + p^2 - 2\vec{q} \cdot \vec{p} + (2\vec{q} \cdot \vec{p} - p^2)x - m^2 + i\epsilon \quad (73)$$

$$D = q^2 - 2(1-x)\vec{q} \cdot \vec{p} + (1-x)p^2 - m^2 + i\epsilon \quad (74)$$

10.2. Complete the Square

$$ax^2 + bx + c = a \left(x + \frac{b}{2a} \right)^2 - \frac{b^2}{4a} + c \quad (75)$$

$$D = [\vec{q} - (1-x)\vec{p}]^2 - (1-x)^2 p^2 + (1-x)p^2 - m^2 + i\epsilon \quad (76)$$

$$D = [\vec{q} - (1-x)\vec{p}]^2 - m^2 + (1-x)xp^2 + i\epsilon \quad (77)$$

$$\Delta = m^2 - (1-x)xp^2 \quad (78)$$

$$D = [\vec{q} - (1-x)\vec{p}]^2 - \Delta + i\epsilon \quad (79)$$

$$\heartsuit = \frac{g^2}{4\pi^4} \int_0^1 dx \int d^4q \frac{\vec{q} \cdot \vec{p} - q^2 - m^2}{([\vec{q} - (1-x)\vec{p}]^2 - \Delta + i\epsilon)^2} \quad (80)$$

10.3. Shift Integrand

$$\vec{k} = \vec{q} - (1-x)\vec{p}, \quad \vec{q} = \vec{k} + (1-x)\vec{p}, \quad d^4k = d^4q \quad (81)$$

$$\heartsuit = \frac{g^2}{4\pi^4} \int_0^1 dx \int d^4k \frac{[\vec{k} + (1-x)\vec{p}] \cdot \vec{p} - [\vec{k} + (1-x)\vec{p}]^2 - m^2}{(k^2 - \Delta + i\epsilon)^2} \quad (82)$$

$$\heartsuit = \frac{g^2}{4\pi^4} \int_0^1 dx \int d^4 q \frac{[\vec{q} + (1-x)\vec{p}] \cdot \vec{p} - [\vec{q} + (1-x)\vec{p}]^2 - m^2}{(q^2 - \Delta + i\epsilon)^2} \quad (83)$$

$$\Delta = m^2 - (1-x)xp^2 \quad (84)$$

$$\heartsuit = \frac{g^2}{4\pi^4} \int_0^1 dx \int d^4 q \frac{(2x-1)\vec{q} \cdot \vec{p} - q^2 - \Delta}{(q^2 - \Delta + i\epsilon)^2} \quad (85)$$

$$\heartsuit = -\frac{g^2}{4\pi^4} \int_0^1 dx \int d^4 q \frac{q^2 + \Delta}{(q^2 - \Delta + i\epsilon)^2} \quad (86)$$

$$\mathbb{I} = \int \frac{d^4 q}{(q^2 - \Delta + i\epsilon)^2}, \quad \mathcal{J} = \int d^4 q \frac{q^2}{(q^2 - \Delta + i\epsilon)^2} \quad (87)$$

$$\heartsuit = -\frac{g^2}{4\pi^4} \int_0^1 dx (\Delta \mathbb{I} + \mathcal{J}) \quad (88)$$

10.4. Divergence

$$\mathbb{I} \sim \int_1^\Lambda dq \frac{q^3}{q^4} = \int_1^\Lambda \frac{dq}{q} = \ln \Lambda, \quad \Lambda \rightarrow \infty \quad (89)$$

$$\mathcal{J} \sim \int_0^\Lambda dq \frac{q^5}{q^4} = \int_0^\Lambda dq q \sim \Lambda^2 \quad (90)$$

10.5. DimReg

$$\mathbb{I}(d) = \int \frac{d^d q}{(q^2 - \Delta + i\epsilon)^2}, \quad \mathcal{J}(d) = \int d^d q \frac{q^2}{(q^2 - \Delta + i\epsilon)^2} \quad (91)$$

$$\int \frac{d^d p}{(p^2 - \Delta + i\epsilon)^2} = \frac{i\pi^{d/2}}{\Delta^{2-d/2}} \Gamma\left(\frac{4-d}{2}\right) \quad (92)$$

$$\int d^d p \frac{p^2}{(p^2 - \Delta + i\epsilon)^2} = \frac{d\pi^{d/2}}{2i\Delta^{1-d/2}} \Gamma\left(\frac{2-d}{2}\right)$$

$$\mathbb{I}(d) = \frac{i\pi^{d/2}}{\Delta^{2-d/2}} \Gamma\left(\frac{4-d}{2}\right), \quad \mathcal{J}(d) = \frac{d\pi^{d/2}}{2i\Delta^{1-d/2}} \Gamma\left(\frac{2-d}{2}\right) \quad (93)$$

$$\mathbb{I}(d) = \frac{i}{\Delta^2} (\pi\Delta)^{d/2} \Gamma\left(2 - \frac{d}{2}\right) \quad (94)$$

$$\mathbb{I}(d) = \frac{i}{\Delta^2} (\pi\Delta)^{d/2} \Gamma\left(1 + 1 - \frac{d}{2}\right) \quad (95)$$

$$\Gamma(x+1) = x\Gamma(x) \quad (96)$$

$$\mathbb{I}(d) = \frac{i}{\Delta^2} (\pi\Delta)^{d/2} \left(1 - \frac{d}{2}\right) \Gamma\left(1 - \frac{d}{2}\right) \quad (97)$$

$$\mathcal{J}(d) = \frac{d(\pi\Delta)^{d/2}}{2i\Delta} \Gamma\left(1 - \frac{d}{2}\right) \quad (98)$$

$$\heartsuit(d) = -\frac{\beta^{4-d} g^2}{4\pi^4} \int_0^1 dx [\Delta \mathbb{I}(d) + \mathcal{J}(d)] \quad (99)$$

$$\heartsuit(d) = i \frac{\beta^{4-d} g^2}{4\pi^4} (d-1) \Gamma\left(1 - \frac{d}{2}\right) \int_0^1 dx \frac{(\pi\Delta)^{d/2}}{\Delta} \quad (100)$$

10.6. Return to 4D

$$\heartsuit = \heartsuit(4) = \heartsuit(4 - \epsilon), \quad \epsilon \rightarrow 0 \quad (101)$$

$$\heartsuit = i \frac{\beta^\epsilon g^2}{4\pi^4} (3 - \epsilon) \Gamma\left(\frac{\epsilon}{2} - 1\right) \int_0^1 dx \frac{(\pi\Delta)^{2-\epsilon/2}}{\Delta} \quad (102)$$

$$\heartsuit = i \frac{g^2}{4\pi^2} (3 - \epsilon) \Gamma\left(\frac{\epsilon}{2} - 1\right) \int_0^1 dx \Delta \left(\frac{\beta}{\sqrt{\pi\Delta}}\right)^\epsilon \quad (103)$$

$$\heartsuit = \frac{ig^2}{2\pi^2} \Gamma\left(\frac{\epsilon}{2}\right) \frac{3 - \epsilon}{\epsilon - 2} \int_0^1 dx \Delta \left(\frac{\beta}{\sqrt{\pi\Delta}}\right)^\epsilon \quad (104)$$

$$\heartsuit = \frac{ig^2}{2\pi^2} \int_0^1 dx \Delta \left(\frac{\beta}{\sqrt{\pi\Delta}}\right)^\epsilon \frac{3 - \epsilon}{\epsilon - 2} \Gamma\left(\frac{\epsilon}{2}\right) \quad (105)$$

$$\heartsuit = \frac{ig^2}{2\pi^2} \int_0^1 dx \Delta \left[-\frac{3}{\epsilon} + \frac{3\gamma - 1}{2} + \frac{3}{2} \ln\left(\frac{\pi\Delta}{\beta^2}\right) \right] \quad (106)$$

$$\heartsuit = \frac{ig^2}{2\pi^2} \left[\left(-\frac{3}{\epsilon} + \frac{3\gamma - 1}{2} \right) \int_0^1 dx \Delta + \frac{3}{2} \int_0^1 dx \Delta \ln\left(\frac{\pi\Delta}{\beta^2}\right) \right] \quad (107)$$

$$\int_0^1 dx \Delta = m^2 - \frac{p^2}{6}, \quad I_1 = \int_0^1 dx \Delta \ln\left(\frac{\pi\Delta}{\beta^2}\right) \quad (108)$$



FIG. 11. ♠

$$\heartsuit = \frac{ig^2}{2\pi^2} \left[\left(-\frac{3}{\epsilon} + \frac{3\gamma-1}{2} \right) \left(m^2 - \frac{p^2}{6} \right) + \frac{3}{2} I_1 \right] \quad (109)$$

$$\heartsuit = \frac{ig^2}{2\pi^2} \left[-\frac{3}{\epsilon} \left(m^2 - \frac{p^2}{6} \right) + \frac{3\gamma-1}{2} \left(m^2 - \frac{p^2}{6} \right) + \frac{3}{2} I_1 \right] \quad (110)$$

$$\heartsuit = \frac{ig^2}{4\pi^2\epsilon} (p^2 - 6m^2) + i\mathbb{F} \quad (111)$$

$$\mathbb{F} = \frac{g^2}{4\pi^2} \left[(3\gamma-1) \left(m^2 - \frac{p^2}{6} \right) + 3I_1 \right] \quad (112)$$

$\mathbb{F}$ is finite.

11. SCATTERING: $\phi \rightarrow \phi$

To extract counterterms I need Green's functions or scattering processes. Consider off-shell ϕ to ϕ scattering. If the scattering is on-shell $p^2 = \mu^2$ and I might not get both $(\delta_\mu$ and $\delta_3)$ counterterms. The trivial part of the S-matrix is free propagation (Fig. 5). To leading order the non-trivial part is

$$\mathcal{M} = \heartsuit + \spadesuit, \quad (113)$$

with $\spadesuit$ in Fig. 11.

$$\spadesuit = i[\delta_3 p^2 - (\delta_\mu + \delta_3)\mu^2] \quad (114)$$

$$\mathcal{M} = \frac{ig^2}{4\pi^2\epsilon} (p^2 - 6m^2) + i\mathbb{F} + i[\delta_3 p^2 - (\delta_\mu + \delta_3)\mu^2] \quad (115)$$

$$\mathcal{M} \propto \frac{g^2}{4\pi^2\epsilon} (p^2 - 6m^2) + \mathbb{F} + \delta_3 p^2 - (\delta_\mu + \delta_3)\mu^2 \quad (116)$$

$$\mathcal{M} \propto \left(\frac{g^2}{4\pi^2\epsilon} + \delta_3 \right) p^2 - \frac{3g^2}{2\pi^2\epsilon} m^2 + \mathbb{F} - (\delta_\mu + \delta_3)\mu^2 \quad (117)$$

I use minimal subtraction.

$$\delta_3 = -\frac{g^2}{4\pi^2\epsilon} \quad (118)$$

$$\mathcal{M} \propto \mathbb{F} - \left(\delta_\mu - \frac{g^2}{4\pi^2\epsilon} \right) \mu^2 - \frac{3g^2}{2\pi^2\epsilon} m^2 \quad (119)$$

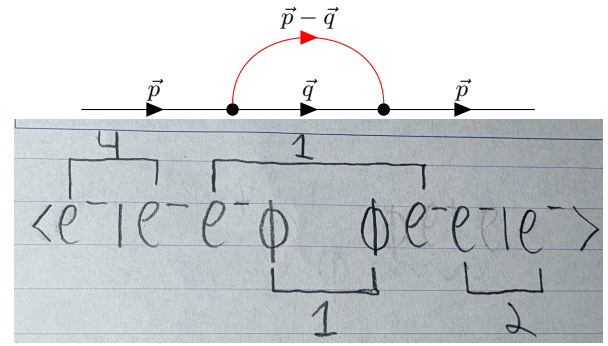


FIG. 12. ♠

This implies

$$\left(\delta_\mu - \frac{g^2}{4\pi^2\epsilon} \right) \mu^2 + \frac{3g^2}{2\pi^2\epsilon} m^2 = 0. \quad (120)$$

$$\delta_\mu = \frac{g^2}{4\pi^2\epsilon} - \frac{3g^2}{2\pi^2\epsilon} \frac{m^2}{\mu^2} \quad (121)$$

$$\delta_\mu = \frac{g^2}{4\pi^2\epsilon} \left(1 - 6 \frac{m^2}{\mu^2} \right) \quad (122)$$

12. ψ SELF-ENERGY

Diagram ♠ in Fig. 12. External spinors are off-shell.

$$\spadesuit = \frac{S}{(2\pi)^4} \int d^4q \frac{i}{(\vec{p}-\vec{q})^2 - \mu^2 + i\epsilon} \frac{(-ig)i \not{q} - m}{q^2 - m^2 + i\epsilon} (-ig) \quad (123)$$

$$S = \frac{4 \times 2}{(2!)^2 2!} = 1 \quad (124)$$

$$\spadesuit = \frac{g^2}{(2\pi)^4} \int d^4q \frac{\not{q} - m}{[(\vec{p}-\vec{q})^2 - \mu^2 + i\epsilon](q^2 - m^2 + i\epsilon)} \quad (125)$$

12.1. Feynman Parameters

$$A = q^2 - m^2 + i\epsilon, \quad B = (\vec{p}-\vec{q})^2 - \mu^2 + i\epsilon \quad (126)$$

$$\spadesuit = \frac{g^2}{(2\pi)^4} \int d^4q \frac{\not{q} - m}{AB} \quad (127)$$

$$\frac{1}{AB} = \int_0^1 \frac{dx}{D^2}, \quad D = A + (B - A)x \quad (128)$$

$$\spadesuit = \frac{g^2}{(2\pi)^4} \int d^4q (q - m) \int_0^1 \frac{dx}{D^2} \quad (129)$$

$$\spadesuit = \frac{g^2}{(2\pi)^4} \int_0^1 dx \int d^4q \frac{q - m}{D^2} \quad (130)$$

$$D = q^2 + [(\vec{p} - \vec{q})^2 - q^2 - \mu^2 + m^2]x - m^2 + i\epsilon \quad (131)$$

$$D = q^2 + (p^2 - 2\vec{q} \cdot \vec{p} - \mu^2 + m^2)x - m^2 + i\epsilon \quad (132)$$

$$D = q^2 - 2x\vec{q} \cdot \vec{p} + (p^2 - \mu^2 + m^2)x - m^2 + i\epsilon \quad (133)$$

12.2. Complete the Square

$$D = (\vec{q} - x\vec{p})^2 - m^2 + [(1-x)p^2 - \mu^2 + m^2]x + i\epsilon \quad (134)$$

$$\Upsilon = m^2 - [(1-x)p^2 - \mu^2 + m^2]x \quad (135)$$

$$D = (\vec{q} - x\vec{p})^2 - \Upsilon + i\epsilon \quad (136)$$

$$\spadesuit = \frac{g^2}{(2\pi)^4} \int_0^1 dx \int d^4q \frac{q - m}{[(\vec{q} - x\vec{p})^2 - \Upsilon + i\epsilon]^2} \quad (137)$$

12.3. Shift Integrand

$$\vec{k} = \vec{q} - x\vec{p}, \quad \vec{q} = \vec{k} + x\vec{p}, \quad d^4k = d^4q \quad (138)$$

$$\spadesuit = \frac{g^2}{(2\pi)^4} \int_0^1 dx \int d^4k \frac{k + x\vec{p} - m}{(k^2 - \Upsilon + i\epsilon)^2} \quad (139)$$

$$\int d^4k \frac{k}{(k^2 - \Upsilon + i\epsilon)^2} = 0 \quad (140)$$

$$\spadesuit = \frac{g^2}{(2\pi)^4} \int_0^1 dx \int d^4k \frac{x\vec{p} - m}{(k^2 - \Upsilon + i\epsilon)^2} \quad (141)$$

$$\spadesuit = \frac{g^2}{(2\pi)^4} \int_0^1 dx (x\vec{p} - m) \int \frac{d^4k}{(k^2 - \Upsilon + i\epsilon)^2} \quad (142)$$

$$\mathbb{I} = \int \frac{d^4k}{(k^2 - \Upsilon + i\epsilon)^2} \quad (143)$$

$$\spadesuit = \frac{g^2}{(2\pi)^4} \int_0^1 dx (x\vec{p} - m) \mathbb{I} \quad (144)$$

12.4. Divergence

$$\mathbb{I} \sim \int_1^\Lambda \frac{dk}{k} = \ln \Lambda, \quad \Lambda \rightarrow \infty \quad (145)$$

12.5. DimReg

$$\mathbb{I}(d) = \int \frac{d^d k}{(k^2 - \Upsilon + i\epsilon)^2} \quad (146)$$

$$\int \frac{d^d p}{(p^2 - \Delta + i\epsilon)^2} = \frac{i\pi^{d/2}}{\Delta^{2-d/2}} \Gamma\left(\frac{4-d}{2}\right) \quad (147)$$

$$\mathbb{I}(d) = \frac{i\pi^{d/2}}{\Upsilon^{2-d/2}} \Gamma\left(\frac{4-d}{2}\right) \quad (148)$$

$$\spadesuit(d) = \frac{\beta^{4-d} g^2}{(2\pi)^4} \int_0^1 dx (x\vec{p} - m) \mathbb{I}(d) \quad (149)$$

$$\spadesuit(d) = \frac{i\pi^{d/2} \beta^{4-d} g^2}{(2\pi)^4} \Gamma\left(\frac{4-d}{2}\right) \int_0^1 dx \frac{x\vec{p} - m}{\Upsilon^{2-d/2}} \quad (150)$$

12.6. Return to 4D

$$\spadesuit = \spadesuit(4) = \spadesuit(4 - \epsilon), \quad \epsilon \rightarrow 0 \quad (151)$$

$$\spadesuit = \frac{i\beta^\epsilon g^2}{16\pi^2 \pi^{\epsilon/2}} \Gamma\left(\frac{\epsilon}{2}\right) \int_0^1 dx \frac{x\vec{p} - m}{\Upsilon^{\epsilon/2}} \quad (152)$$

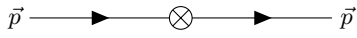
$$\spadesuit = \frac{ig^2}{16\pi^2} \int_0^1 dx (x\vec{p} - m) \left(\frac{\beta}{\sqrt{\pi\Upsilon}}\right)^\epsilon \Gamma\left(\frac{\epsilon}{2}\right) \quad (153)$$

$$\spadesuit = \frac{ig^2}{16\pi^2} \int_0^1 dx (x\vec{p} - m) \left[\frac{2}{\epsilon} - \gamma + \ln\left(\frac{\beta^2}{\pi\Upsilon}\right)\right] \quad (154)$$

$$\spadesuit = \frac{ig^2}{16\pi^2} \left[\left(\frac{2}{\epsilon} - \gamma\right) \int_0^1 dx (x\vec{p} - m) + I_2\right] \quad (155)$$

$$I_2 = \int_0^1 dx (x\vec{p} - m) \ln\left(\frac{\beta^2}{\pi\Upsilon}\right) \quad (156)$$

$$\int_0^1 dx (x\vec{p} - m) = \frac{\vec{p}}{2} - m \quad (157)$$

FIG. 13. $\mathfrak{V}$

$$\spadesuit = \frac{ig^2}{16\pi^2} \left[\left(\frac{2}{\epsilon} - \gamma \right) \left(\frac{\not{p}}{2} - m \right) + I_2 \right] \quad (158)$$

$$\spadesuit = \frac{ig^2}{16\pi^2\epsilon} (\not{p} - 2m) + i\mathcal{F} \quad (159)$$

$$\mathcal{F} = \frac{g^2}{16\pi^2} \left[I_2 - \gamma \left(\frac{\not{p}}{2} - m \right) \right] \quad (160)$$

13. SCATTERING: $\psi \rightarrow \psi$

For off-shell scattering the leading contribution is

$$\mathcal{M} = \spadesuit + \mathfrak{V}, \quad (161)$$

with $\mathfrak{V}$ in Fig. 13.

$$\mathfrak{V} = i(\delta_2 \not{p} + \delta_4 m) \quad (162)$$

$$\mathcal{M} = \frac{ig^2}{16\pi^2\epsilon} (\not{p} - 2m) + i\mathcal{F} + i(\delta_2 \not{p} + \delta_4 m) \quad (163)$$

$$\mathcal{M} \propto \frac{g^2}{16\pi^2\epsilon} (\not{p} - 2m) + \mathcal{F} + \delta_2 \not{p} + \delta_4 m \quad (164)$$

$$\mathcal{M} \propto \left(\frac{g^2}{16\pi^2\epsilon} + \delta_2 \right) \not{p} + \left(\delta_4 - \frac{g^2}{8\pi^2\epsilon} \right) m + \mathcal{F} \quad (165)$$

This implies

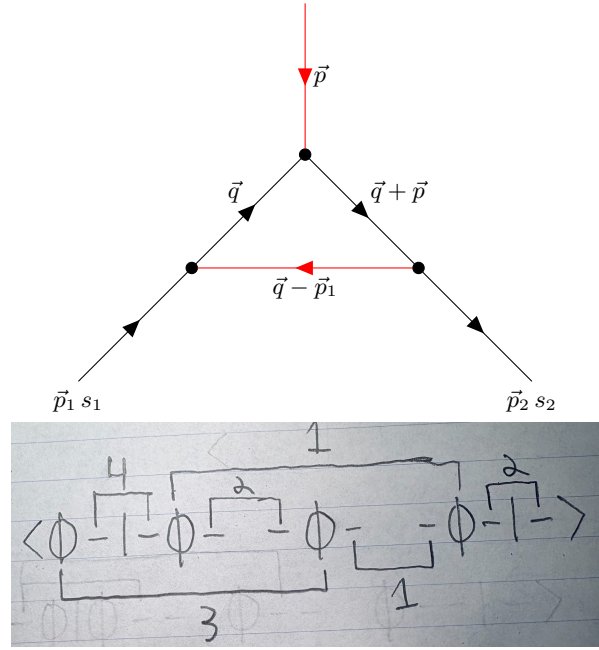
$$\delta_2 = -\frac{g^2}{16\pi^2\epsilon}, \quad \delta_4 = \frac{g^2}{8\pi^2\epsilon}. \quad (166)$$

14. VERTEX LOOP

The diagram is in Fig. 14. This time the external fields are on-shell. Conservation of momentum implies

$$\vec{p} + \vec{p}_1 = \vec{p}_2. \quad (167)$$

$$\clubsuit = \frac{S}{(2\pi)^4} \int d^4q \frac{i}{(\vec{q} - \vec{p}_1)^2 - \mu^2 + i\epsilon} \bar{u}_{s_2}(\vec{p}_2) (-ig) \frac{\not{q} + \not{p} - m}{(\vec{q} + \vec{p})^2 - m^2 + i\epsilon} (-ig) i \frac{\not{q} - m}{q^2 - m^2 + i\epsilon} (-ig) u_{s_1}(\vec{p}_1) \quad (168)$$

FIG. 14. $\clubsuit$

$$S = \frac{4 \times 3 \times 2^2}{(2!)^3 3!} = 1, \quad u_{s_n}(\vec{p}_n) = u_n \quad (169)$$

$$\clubsuit = \frac{i^3 (-ig)^3}{(2\pi)^4} \int \frac{d^4q}{(\vec{q} - \vec{p}_1)^2 - \mu^2 + i\epsilon} \bar{u}_2 \frac{\not{q} + \not{p} - m}{(\vec{q} + \vec{p})^2 - m^2 + i\epsilon} \frac{\not{q} - m}{q^2 - m^2 + i\epsilon} u_1 \quad (170)$$

$$\clubsuit = \frac{g^3}{(2\pi)^4} \int \frac{d^4q}{(\vec{q} - \vec{p}_1)^2 - \mu^2 + i\epsilon} \frac{N}{[(\vec{q} + \vec{p})^2 - m^2 + i\epsilon](q^2 - m^2 + i\epsilon)} \quad (171)$$

$$N = \bar{u}_2 (\not{q} + \not{p} - m) (\not{q} - m) u_1 \quad (172)$$

14.1. Kinematics

$$p^2 = \mu^2, \quad p_1^2 = p_2^2 = m^2 \quad (173)$$

$$\vec{p} + \vec{p}_1 = \vec{p}_2 \implies (\vec{p} + \vec{p}_1)^2 = p_2^2 \implies p^2 + p_1^2 + 2\vec{p} \cdot \vec{p}_1 = p_2^2 \implies 2\vec{p} \cdot \vec{p}_1 = -\mu^2 \quad (174)$$

14.2. Dirac Algebra

$$\vec{p} = \vec{p}_2 - \vec{p}_1 \quad (175)$$

$$N = \bar{u}_2(\not{q} + \not{p}_2 - \not{p}_1 - m)(\not{q} - m)u_1 \quad (176)$$

$$\not{p}_1 u_1 = m u_1, \quad \bar{u}_2 \not{p}_2 = m \bar{u}_2 \quad (177)$$

$$N = \bar{u}_2(\not{q} - \not{p}_1)(\not{q} - m)u_1 \quad (178)$$

$$N = \bar{u}_2(\not{q}^2 - m\not{q} - \not{p}_1\not{q} + m\not{p}_1)u_1 \quad (179)$$

In Sec. 20.2 I show

$$\not{p}^2 = p^2. \quad (180)$$

$$N = \bar{u}_2(q^2 - m\not{q} - \not{p}_1\not{q} + m\not{p}_1)u_1 \quad (181)$$

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} \implies \gamma^\mu\gamma^\nu = 2\eta^{\mu\nu} - \gamma^\nu\gamma^\mu \quad (182)$$

$$\not{q}\not{p} = q_\alpha p_\beta \gamma^\alpha \gamma^\beta = q_\alpha p_\beta (2\eta^{\alpha\beta} - \gamma^\beta \gamma^\alpha) = 2\vec{q} \cdot \vec{p} - \not{p}\not{q} \quad (183)$$

$$N = \bar{u}_2(q^2 - m\not{q} - 2\vec{q} \cdot \vec{p}_1 + \not{q}\not{p}_1 + m\not{p}_1)u_1 \quad (184)$$

$$N = \bar{u}_2(q^2 - 2\vec{q} \cdot \vec{p}_1 + m^2)u_1 \quad (185)$$

$$N = \bar{u}_2 u_1 (q^2 - 2\vec{q} \cdot \vec{p}_1 + m^2) \quad (186)$$

$$N = \mathfrak{B}(q^2 - 2\vec{q} \cdot \vec{p}_1 + m^2), \quad \mathfrak{B} = \bar{u}_2 u_1 \quad (187)$$

$$\clubsuit = \frac{g^3 \mathfrak{B}}{(2\pi)^4} \int \frac{d^4 q}{(\vec{q} - \vec{p}_1)^2 - \mu^2 + i\epsilon} \frac{q^2 - 2\vec{q} \cdot \vec{p}_1 + m^2}{[(\vec{q} + \vec{p})^2 - m^2 + i\epsilon](q^2 - m^2 + i\epsilon)} \quad (188)$$

14.3. Feynman Parameters

$$A = q^2 - m^2 + i\epsilon, \quad B = (\vec{q} + \vec{p})^2 - m^2 + i\epsilon, \quad C = (\vec{q} - \vec{p}_1)^2 - \mu^2 + i\epsilon \quad (189)$$

$$\clubsuit = \frac{g^3 \mathfrak{B}}{(2\pi)^4} \int d^4 q \frac{q^2 - 2\vec{q} \cdot \vec{p}_1 + m^2}{ABC} \quad (190)$$

$$\frac{1}{ABC} = 2 \int_0^1 dx dy dz \frac{\delta(x+y+z-1)}{D^3}, \quad D = xA + yB + zC \quad (191)$$

$$\clubsuit = \frac{g^3 \mathfrak{B}}{8\pi^4} \int_0^1 dx dy dz \delta(x+y+z-1) \int d^4 q \frac{q^2 - 2\vec{q} \cdot \vec{p}_1 + m^2}{D^3} \quad (192)$$

$$D = x(q^2 - m^2 + i\epsilon) + y(q^2 + p^2 + 2\vec{q} \cdot \vec{p} - m^2 + i\epsilon) + z(q^2 + p_1^2 - 2\vec{q} \cdot \vec{p}_1 - \mu^2 + i\epsilon) \quad (193)$$

$$p^2 = \mu^2, \quad p_1^2 = m^2 \quad (194)$$

$$D = x(q^2 - m^2 + i\epsilon) + y(q^2 + 2\vec{q} \cdot \vec{p} + \mu^2 - m^2 + i\epsilon) + z(q^2 - 2\vec{q} \cdot \vec{p}_1 - \mu^2 + m^2 + i\epsilon) \quad (195)$$

$$D = (x+y+z)q^2 + 2\vec{q} \cdot (y\vec{p} - z\vec{p}_1) + (y-z)\mu^2 + (z-x-y)m^2 + (x+y+z)i\epsilon \quad (196)$$

The delta function enforces $x+y+z=1$.

$$D = q^2 + 2\vec{q} \cdot (y\vec{p} - z\vec{p}_1) + (y-z)\mu^2 + (z-x-y)m^2 + i\epsilon \quad (197)$$

$$D = q^2 + 2\vec{q} \cdot (y\vec{p} - z\vec{p}_1) + (y-z)\mu^2 + (2z-1)m^2 + i\epsilon \quad (198)$$

14.4. Complete the Square

$$ax^2 + bx + c = a \left(x + \frac{b}{2a} \right)^2 - \frac{b^2}{4a} + c \quad (199)$$

$$D = (\vec{q} + y\vec{p} - z\vec{p}_1)^2 - (y\vec{p} - z\vec{p}_1)^2 + (y-z)\mu^2 + (2z-1)m^2 + i\epsilon \quad (200)$$

$$D = (\vec{q} + y\vec{p} - z\vec{p}_1)^2 - y^2 p^2 - z^2 p_1^2 + 2yz\vec{p} \cdot \vec{p}_1 + (y-z)\mu^2 + (2z-1)m^2 + i\epsilon \quad (201)$$

$$p^2 = \mu^2, \quad p_1^2 = m^2, \quad 2\vec{p} \cdot \vec{p}_1 = -\mu^2 \quad (202)$$

$$D = (\vec{q} + y\vec{p} - z\vec{p}_1)^2 + (y-z-y^2-yz)\mu^2 + (2z-1-z^2)m^2 + i\epsilon \quad (203)$$

$$y-z-y^2-yz = y(1-y-z) - z = xy - z \quad (204)$$

$$D = (\vec{q} + y\vec{p} - z\vec{p}_1)^2 + (xy - z)\mu^2 - (1-z)^2 m^2 + i\epsilon \quad (205)$$

$$D = (\vec{q} + y\vec{p} - z\vec{p}_1)^2 - \mathfrak{U} + i\epsilon \quad (206)$$

$$\mathfrak{U} = (1-z)^2 m^2 - (xy - z)\mu^2 \quad (207)$$

$$\clubsuit = \frac{g^3 \mathfrak{B}}{8\pi^4} \int_0^1 dx dy dz \delta(x+y+z-1) \int d^4 q \frac{q^2 - 2\vec{q} \cdot \vec{p}_1 + m^2}{[(\vec{q} + y\vec{p} - z\vec{p}_1)^2 - \mathfrak{U} + i\epsilon]^3} \quad (208)$$

14.5. Shift Integrand

$$\vec{k} = \vec{q} + y\vec{p} - z\vec{p}_1, \quad \vec{q} = \vec{k} + z\vec{p}_1 - y\vec{p}, \quad d^4k = d^4q \quad (209)$$

$$\clubsuit = \frac{g^3 \mathfrak{B}}{8\pi^4} \int_0^1 dx dy dz \delta(x+y+z-1) \int d^4k \frac{(\vec{k} - y\vec{p} + z\vec{p}_1)^2 - 2(\vec{k} - y\vec{p} + z\vec{p}_1) \cdot \vec{p}_1 + m^2}{(k^2 - \mathfrak{U} + i\epsilon)^3} \quad (210)$$

$$\clubsuit = \frac{g^3 \mathfrak{B}}{8\pi^4} \int_0^1 dx dy dz \delta(x+y+z-1) \int d^4k \frac{(\vec{k} - y\vec{p} + z\vec{p}_1)^2 - \cancel{2\vec{k} \cdot \vec{p}_1} + 2y\vec{p} \cdot \vec{p}_1 - 2zp_1^2 + m^2}{(k^2 - \mathfrak{U} + i\epsilon)^3} \quad (211)$$

$$p^2 = \mu^2, \quad p_1^2 = m^2, \quad 2\vec{p} \cdot \vec{p}_1 = -\mu^2 \quad (212)$$

$$\clubsuit = \frac{g^3 \mathfrak{B}}{8\pi^4} \int_0^1 dx dy dz \delta(x+y+z-1) \int \frac{d^4k}{(k^2 - \mathfrak{U} + i\epsilon)^3} \frac{1}{[(\vec{k} - y\vec{p} + z\vec{p}_1)^2 - y\mu^2 + (1-2z)m^2]} \quad (213)$$

$$\clubsuit = \frac{g^3 \mathfrak{B}}{8\pi^4} \int_0^1 dx dy dz \delta(x+y+z-1) \int \frac{d^4k}{(k^2 - \mathfrak{U} + i\epsilon)^3} \frac{1}{[k^2 + y^2p^2 + z^2p_1^2 + \cancel{2\vec{k} \cdot (z\vec{p}_1 - y\vec{p})} - 2yz\vec{p} \cdot \vec{p}_1 - y\mu^2 + (1-2z)m^2]} \quad (214)$$

$$\clubsuit = \frac{g^3 \mathfrak{B}}{8\pi^4} \int_0^1 dx dy dz \delta(x+y+z-1) \int d^4k \frac{k^2 + y^2p^2 + z^2p_1^2 - 2yz\vec{p} \cdot \vec{p}_1 - y\mu^2 + (1-2z)m^2}{(k^2 - \mathfrak{U} + i\epsilon)^3} \quad (215)$$

$$\clubsuit = \frac{g^3 \mathfrak{B}}{8\pi^4} \int_0^1 dx dy dz \delta(x+y+z-1) \int d^4k \frac{k^2 + y(y+z-1)\mu^2 + (1-2z+z^2)m^2}{(k^2 - \mathfrak{U} + i\epsilon)^3} \quad (216)$$

$$\clubsuit = \frac{g^3 \mathfrak{B}}{8\pi^4} \int_0^1 dx dy dz \delta(x+y+z-1) \int d^4k \frac{k^2 - xy\mu^2 + (1-z)^2m^2}{(k^2 - \mathfrak{U} + i\epsilon)^3} \quad (217)$$

$$\mathbb{I} = \int \frac{d^4k}{(k^2 - \mathfrak{U} + i\epsilon)^3}, \quad \mathscr{J} = \int d^4k \frac{k^2}{(k^2 - \mathfrak{U} + i\epsilon)^3} \quad (218)$$

$$\clubsuit = \frac{g^3 \mathfrak{B}}{8\pi^4} \int_0^1 dx dy dz \delta(x+y+z-1) (\mathscr{J} + [(1-z)^2m^2 - xy\mu^2]\mathbb{I}) \quad (219)$$

14.6. Convergent Integral

$$\mathbb{I} \sim \int^\Lambda dk \frac{k^3}{k^6} = \int^\Lambda \frac{dk}{k^3} \sim \frac{1}{\Lambda^2}, \quad \Lambda \rightarrow \infty \quad (220)$$

$$i\mathfrak{F} = \frac{g^3}{8\pi^4} \int_0^1 dx dy dz \delta(x+y+z-1) [(1-z)^2m^2 - xy\mu^2]\mathbb{I} \quad (221)$$

$\mathfrak{F}$ is finite.

$$\clubsuit = i\mathfrak{B}\mathfrak{F} + \frac{g^3 \mathfrak{B}}{8\pi^4} \int_0^1 dx dy dz \delta(x+y+z-1) \mathscr{J} \quad (222)$$

14.7. Divergent Integral

$$\mathscr{J} \sim \int_1^\Lambda dk \frac{k^5}{k^6} = \int_1^\Lambda \frac{dk}{k} = \ln \Lambda, \quad \Lambda \rightarrow \infty \quad (223)$$

14.8. DimReg

$$T = g^2 \mathscr{J} = g^2 \int d^4k \frac{k^2}{(k^2 - \mathfrak{U} + i\epsilon)^3} \quad (224)$$

$$\clubsuit = i\mathfrak{B}\mathfrak{F} + \frac{g^3 \mathfrak{B}}{8\pi^4} \int_0^1 dx dy dz \delta(x+y+z-1) T \quad (225)$$

$$T(d) = g^2 \beta^{4-d} \int d^d k \frac{k^2}{(k^2 - \mathfrak{U} + i\epsilon)^3} \quad (226)$$

From Schwartz Eq. B.35:

$$\int \frac{d^d k}{(2\pi)^d} \frac{k^{2a}}{(k^2 - \Delta + i\epsilon)^b} = \frac{i(-1)^{a-b}}{(4\pi)^{d/2} \Delta^{b-a-d/2}} \frac{\Gamma(a+d/2)\Gamma(b-a-d/2)}{\Gamma(b)\Gamma(d/2)}. \quad (227)$$

$$\int d^d k \frac{k^2}{(k^2 - \mathfrak{U} + i\epsilon)^3} = \frac{i(-1)^{-2}(2\pi)^d}{(4\pi)^{d/2} \mathfrak{U}^{2-d/2}} \frac{\Gamma(1+d/2)\Gamma(2-d/2)}{\Gamma(3)\Gamma(d/2)} \quad (228)$$

$$\int d^d k \frac{k^2}{(k^2 - \mathfrak{U} + i\epsilon)^3} = \frac{i\pi^{d/2}}{2\mathfrak{U}^{2-d/2}} \frac{\Gamma(1+d/2)\Gamma(2-d/2)}{\Gamma(d/2)} \quad (229)$$

$$\frac{\Gamma(1+d/2)\Gamma(2-d/2)}{\Gamma(d/2)} = \frac{d}{2}\Gamma\left(\frac{4-d}{2}\right) \quad (230)$$

$$\int d^d k \frac{k^2}{(k^2 - \mathcal{U} + i\epsilon)^3} = \frac{id\pi^{d/2}}{4\mathcal{U}^{2-d/2}}\Gamma\left(\frac{4-d}{2}\right) \quad (231)$$

$$T(d) = \frac{ig^2 d\pi^{d/2} \beta^{4-d}}{4\mathcal{U}^{2-d/2}}\Gamma\left(\frac{4-d}{2}\right) \quad (232)$$

14.9. Return to 4D

$$T = T(4) = T(4 - \epsilon), \quad \epsilon \rightarrow 0 \quad (233)$$

$$T = \frac{ig^2\pi^2\beta^\epsilon}{4(\pi\mathcal{U})^{\epsilon/2}}(4 - \epsilon)\Gamma\left(\frac{\epsilon}{2}\right) \quad (234)$$

$$T = \frac{ig^2\pi^2}{4}(4 - \epsilon)\left(\frac{\beta}{\sqrt{\pi\mathcal{U}}}\right)^\epsilon \Gamma\left(\frac{\epsilon}{2}\right) \quad (235)$$

$$f = g^2\pi^2 \left[\ln\left(\frac{\beta^2}{\pi\mathcal{U}}\right) - \frac{1}{2} - \gamma \right] \quad (236)$$

$$T = \frac{2ig^2\pi^2}{\epsilon} + if \quad (237)$$

$$\clubsuit = i\mathfrak{B}\mathfrak{F} + \frac{g^3\mathfrak{B}}{8\pi^4} \int_0^1 dx dy dz \delta(x+y+z-1)T \quad (238)$$

$$F = \frac{g}{8\pi^4} \int_0^1 dx dy dz \delta(x+y+z-1)f \quad (239)$$

$$\clubsuit = i\mathfrak{B}(\mathfrak{F} + F) + \frac{ig^3\mathfrak{B}}{4\pi^2\epsilon} \int_0^1 dx dy dz \delta(x+y+z-1) \quad (240)$$

$$\mathcal{F} = \mathfrak{F} + F \quad (241)$$

$\mathcal{F}$ is finite.

$$\int_0^1 dx dy dz \delta(x+y+z-1) = \frac{1}{2} \quad (242)$$

$$\boxed{\clubsuit = i\mathfrak{B}\mathcal{F} + \frac{ig^3\mathfrak{B}}{8\pi^2\epsilon}} \quad (243)$$

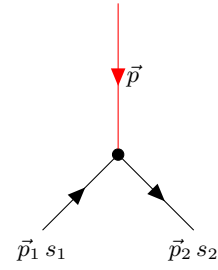


FIG. 15. $\psi \phi \rightarrow \psi$

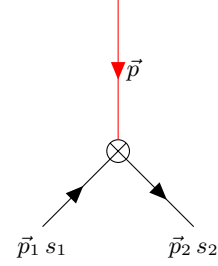


FIG. 16. $\star$

15. SCATTERING: $\psi \phi \rightarrow \psi$

15.1. Leading Order

There is one tree-level diagram (Fig. 15). I don't evaluate it because it won't yield a counterterm.

15.2. Leading Correction

The leading correction is

$$\Delta\mathcal{M} = \clubsuit + \star, \quad (244)$$

with $\star$ in Fig. 16.

$$\star = \bar{u}_2(-i\delta_5 g)u_1 = -i\delta_5 g \bar{u}_2 u_1 = -i\delta_5 g \mathfrak{B} \quad (245)$$

$$\Delta\mathcal{M} = i\mathfrak{B}\mathcal{F} + \frac{ig^3\mathfrak{B}}{8\pi^2\epsilon} - i\delta_5 g \mathfrak{B} \quad (246)$$

$$\Delta\mathcal{M} \propto \mathcal{F} + \frac{g^3}{8\pi^2\epsilon} - \delta_5 g = \mathcal{F} + \left(\frac{g^2}{8\pi^2\epsilon} - \delta_5\right)g \quad (247)$$

This implies

$$\boxed{\delta_5 = \frac{g^2}{8\pi^2\epsilon}}. \quad (248)$$

16. δ_1 & δ_M

$$Z_4 = Z_2 \sqrt{Z_m} \quad (249)$$

$$Z_m = \left(\frac{Z_4}{Z_2} \right)^2 \quad (250)$$

$$Z_i = 1 + \delta_i \quad (251)$$

$$\delta_m = \left(\frac{1 + \delta_4}{1 + \delta_2} \right)^2 - 1 \quad (252)$$

$$\delta_2 = -\frac{g^2}{16\pi^2\epsilon} \quad \delta_4 = \frac{g^2}{8\pi^2\epsilon} \quad (253)$$

I take the $|g| \ll |\epsilon| \ll 1$ limit.

$$\boxed{\delta_m = \frac{3g^2}{8\pi^2\epsilon}} \quad (254)$$

$$Z_5 = Z_1 Z_2 \sqrt{Z_3} \quad (255)$$

$$Z_1 = \frac{Z_5}{Z_2 \sqrt{Z_3}} \quad (256)$$

$$\delta_1 = \frac{1 + \delta_5}{(1 + \delta_2)\sqrt{1 + \delta_3}} - 1 \quad (257)$$

$$\delta_5 = \frac{g^2}{8\pi^2\epsilon}, \quad \delta_3 = -\frac{g^2}{4\pi^2\epsilon} \quad (258)$$

$$\boxed{\delta_1 = \frac{5g^2}{16\pi^2\epsilon}} \quad (259)$$

Part III

Problem 3

17. LAGRANGIAN

The QCD Lagrangian with one quark field and no ghosts (I don't need ghost fields at tree level) in Feynman gauge is

$$\mathcal{L} = \bar{\psi}(i\not{D} - m)\psi - \frac{F^2}{4} - \frac{(\partial A)^2}{2}. \quad (260)$$

$$\partial A^a = \eta^{\mu\nu} \partial_\mu A_\nu^a = \partial^\mu A_\mu^a \quad (261)$$

$$(\partial A)^2 = \sum_{a=1}^8 (\partial A^a)(\partial A^a) = (\partial A^a)(\partial A^a) \quad (262)$$

I make the sum over repeated adjoint indices implicit.

$$F^2 = \sum_{a=1}^8 F_{\mu\nu}^a F^{a\mu\nu} = F_{\mu\nu}^a F^{a\mu\nu} \quad (263)$$

$$\begin{aligned} F_{\mu\nu}^a &= \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g \sum_{bc=1}^8 f^{abc} A_\mu^b A_\nu^c \\ &= \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g f^{abc} A_\mu^b A_\nu^c \end{aligned} \quad (264)$$

$$\begin{aligned} F^2 &= (\partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g f^{abc} A_\mu^b A_\nu^c) \\ &\quad (\partial^\mu A^{a\nu} - \partial^\nu A^{a\mu} + g f^{ade} A^{d\mu} A^{e\nu}) \end{aligned} \quad (265)$$

$$\begin{aligned} F^2 &= (\partial_\mu A_\nu^a)(\partial^\mu A^{a\nu}) - (\partial_\mu A_\nu^a)(\partial^\nu A^{a\mu}) \\ &+ g f^{ade} (\partial_\mu A_\nu^a) A^{d\mu} A^{e\nu} - (\partial_\nu A_\mu^a)(\partial^\mu A^{a\nu}) + (\partial_\nu A_\mu^a)(\partial^\nu A^{a\mu}) \\ &- g f^{ade} (\partial_\nu A_\mu^a) A^{d\mu} A^{e\nu} + g f^{abc} (\partial^\mu A^{a\nu}) A_\mu^b A_\nu^c \\ &- g f^{abc} (\partial^\nu A^{a\mu}) A_\mu^b A_\nu^c + g^2 f^{ade} f^{abc} A_\mu^b A_\nu^c A^{d\mu} A^{e\nu} \end{aligned} \quad (266)$$

$$\begin{aligned} F^2 &= (\partial_\mu A_\nu^a)(\partial^\mu A^{a\nu}) + (\partial_\nu A_\mu^a)(\partial^\nu A^{a\mu}) \\ &- (\partial_\mu A_\nu^a)(\partial^\nu A^{a\mu}) - (\partial_\nu A_\mu^a)(\partial^\mu A^{a\nu}) \\ &+ g f^{ade} (\partial_\mu A_\nu^a) A^{d\mu} A^{e\nu} + g f^{abc} (\partial^\mu A^{a\nu}) A_\mu^b A_\nu^c \\ &- g f^{ade} (\partial_\nu A_\mu^a) A^{d\mu} A^{e\nu} - g f^{abc} (\partial^\nu A^{a\mu}) A_\mu^b A_\nu^c \\ &+ g^2 f^{ade} f^{abc} A_\mu^b A_\nu^c A^{d\mu} A^{e\nu} \end{aligned} \quad (267)$$

$$\begin{aligned} F^2 &= (\partial_\mu A_\nu^a)(\partial^\mu A^{a\nu}) + (\partial_\mu A_\nu^a)(\partial^\mu A^{a\nu}) \\ &- (\partial_\mu A_\nu^a)(\partial^\nu A^{a\mu}) - (\partial_\mu A_\nu^a)(\partial^\nu A^{a\mu}) \\ &+ g f^{abc} (\partial_\mu A_\nu^a) A^{b\mu} A^{c\nu} + g f^{abc} (\partial^\mu A^{a\nu}) A_\mu^b A_\nu^c \\ &- g f^{abc} (\partial_\nu A_\mu^a) A^{b\mu} A^{c\nu} - g f^{abc} (\partial^\nu A^{a\mu}) A_\mu^b A_\nu^c \\ &+ g^2 f^{ade} f^{abc} A_\mu^b A_\nu^c A^{d\mu} A^{e\nu} \end{aligned} \quad (268)$$

$$\begin{aligned} F^2 &= 2(\partial_\mu A_\nu^a)(\partial^\mu A^{a\nu}) - 2(\partial_\mu A_\nu^a)(\partial^\nu A^{a\mu}) \\ &+ g f^{abc} (\partial^\mu A^{a\nu}) A_\mu^b A_\nu^c + g f^{abc} (\partial^\mu A^{a\nu}) A_\mu^b A_\nu^c \\ &- g f^{abc} (\partial^\nu A^{a\mu}) A_\mu^b A_\nu^c - g f^{abc} (\partial^\nu A^{a\mu}) A_\mu^b A_\nu^c \\ &+ g^2 f^{ade} f^{abc} A_\mu^b A_\nu^c A^{d\mu} A^{e\nu} \end{aligned} \quad (269)$$

$$\begin{aligned} F^2 &= 2(\partial_\mu A_\nu^a)(\partial^\mu A^{a\nu}) - 2(\partial_\mu A_\nu^a)(\partial^\nu A^{a\mu}) \\ &+ g f^{abc} [(\partial^\mu A^{a\nu}) A_\mu^b A_\nu^c + (\partial^\mu A^{a\nu}) A_\mu^b A_\nu^c \\ &- (\partial^\nu A^{a\mu}) A_\mu^b A_\nu^c - (\partial^\nu A^{a\mu}) A_\mu^b A_\nu^c] \\ &+ g^2 f^{ade} f^{abc} A_\mu^b A_\nu^c A^{d\mu} A^{e\nu} \end{aligned} \quad (270)$$

$$\begin{aligned} F^2 &= 2(\partial_\mu A_\nu^a)(\partial^\mu A^{a\nu}) - 2(\partial_\mu A_\nu^a)(\partial^\nu A^{a\mu}) \\ &+ 2g f^{abc} [(\partial^\mu A^{a\nu}) A_\mu^b A_\nu^c - (\partial^\nu A^{a\mu}) A_\mu^b A_\nu^c] \\ &+ g^2 f^{ade} f^{abc} A_\mu^b A_\nu^c A^{d\mu} A^{e\nu} \end{aligned} \quad (271)$$

At tree level there is no 4-gluon vertex for the $q\bar{q} \rightarrow gg$ reaction. So I ignore the $AAAA$ term.

$$\begin{aligned} F^2 &= 2(\partial_\mu A_\nu^a)(\partial^\mu A^{a\nu}) - 2(\partial_\mu A_\nu^a)(\partial^\nu A^{a\mu}) \\ &+ 2g f^{abc} (A_\mu^b A_\nu^c \partial^\mu A^{a\nu} - A_\mu^b A_\nu^c \partial^\nu A^{a\mu}) \end{aligned} \quad (272)$$

$$\begin{aligned} F^2 &= 2(\partial_\mu A_\nu^a)(\partial^\mu A^{a\nu}) - 2(\partial_\mu A_\nu^a)(\partial^\nu A^{a\mu}) \\ &+ 2g (f^{abc} A_\mu^b A_\nu^c \partial^\mu A^{a\nu} - f^{acb} A_\mu^b A_\nu^c \partial^\nu A^{a\mu}) \end{aligned} \quad (273)$$

$$f^{abc} = -f^{acb} \quad (274)$$

$$\begin{aligned} F^2 &= 2(\partial_\mu A_\nu^a)(\partial^\mu A^{a\nu}) - 2(\partial_\mu A_\nu^a)(\partial^\nu A^{a\mu}) \\ &+ 2g f^{abc} (A_\mu^b A_\nu^c \partial^\mu A^{a\nu} + A_\nu^b A_\mu^c \partial^\nu A^{a\mu}) \end{aligned} \quad (275)$$

$$\begin{aligned} F^2 &= 2(\partial_\mu A_\nu^a)(\partial^\mu A^{a\nu}) - 2(\partial_\mu A_\nu^a)(\partial^\nu A^{a\mu}) \\ &+ 2g f^{abc} (A_\mu^b A_\nu^c \partial^\mu A^{a\nu} + A_\mu^b A_\nu^c \partial^\mu A^{a\nu}) \end{aligned} \quad (276)$$

$$\begin{aligned} F^2 &= 2(\partial_\mu A_\nu^a)(\partial^\mu A^{a\nu}) - 2(\partial_\mu A_\nu^a)(\partial^\nu A^{a\mu}) \\ &+ 4g f^{abc} A_\mu^b A_\nu^c \partial^\mu A^{a\nu} \end{aligned} \quad (277)$$

$$\begin{aligned} \mathcal{L} &= \bar{\psi}(i\not{D} - m)\psi + \frac{(\partial_\mu A_\nu^a)(\partial^\nu A^{a\mu}) - (\partial_\mu A_\nu^a)(\partial^\mu A^{a\nu})}{2} \\ &- \frac{(\partial A)^2}{2} - g f^{abc} A_\mu^b A_\nu^c \partial^\mu A^{a\nu} \end{aligned} \quad (278)$$

$$\begin{aligned} \mathcal{L} &= i\bar{\psi}\not{D}\psi - m\bar{\psi}\psi - g f^{abc} A_\mu^b A_\nu^c \partial^\mu A^{a\nu} \\ &+ \frac{(\partial_\mu A_\nu^a)(\partial^\nu A^{a\mu}) - (\partial_\mu A_\nu^a)(\partial^\mu A^{a\nu}) - (\partial A)^2}{2} \end{aligned} \quad (279)$$

$$\bar{\psi}\psi = \sum_{i=1}^3 \bar{\psi}_i\psi_i = \bar{\psi}_i\psi_i \quad (280)$$

$$(\partial A_1)^2 = (\partial_\mu A_\nu^a)(\partial^\nu A^{a\mu}), \quad (\partial A_2)^2 = (\partial_\mu A_\nu^a)(\partial^\mu A^{a\nu}) \quad (281)$$

$$\mathcal{L} = i\bar{\psi}\not{D}\psi - m\bar{\psi}\psi - gf^{abc}A_\mu^b A_\nu^c \partial^\mu A^{a\nu} + \frac{(\partial A_1)^2 - (\partial A_2)^2 - (\partial A)^2}{2} \quad (282)$$

$$(dA)^2 = (\partial A_1)^2 - (\partial A_2)^2 - (\partial A)^2 \quad (283)$$

$$\mathcal{L} = i\bar{\psi}\not{D}\psi - m\bar{\psi}\psi + \frac{(dA)^2}{2} - gf^{abc}A_\mu^b A_\nu^c \partial^\mu A^{a\nu} \quad (284)$$

$$\mathcal{L} = i\bar{\psi}\gamma^\mu D_\mu\psi - m\bar{\psi}\psi + \frac{(dA)^2}{2} - gf^{abc}A_\mu^b A_\nu^c \partial^\mu A^{a\nu} \quad (285)$$

$$\begin{aligned} i\bar{\psi}\gamma^\mu D_\mu\psi &= \sum_{ij=1}^3 \bar{\psi}_i(\delta_{ij}i\gamma^\mu\partial_\mu + g\gamma^\mu A_\mu^a T_{ij}^a)\psi_j = \\ \bar{\psi}_i(\delta_{ij}i\gamma^\mu\partial_\mu + g\gamma^\mu A_\mu^a T_{ij}^a)\psi_j &= \bar{\psi}_i(i\delta_{ij}\not{\partial} + g\mathbf{A}^a T_{ij}^a)\psi_j \\ &= i\delta_{ij}\bar{\psi}_i\not{\partial}\psi_j + g\bar{\psi}_i\mathbf{A}^a T_{ij}^a\psi_j = i\bar{\psi}_i\not{\partial}\psi_i + g\bar{\psi}_i\mathbf{A}^a T_{ij}^a\psi_j \\ &= i\bar{\psi}\not{\partial}\psi + g\bar{\psi}_i\mathbf{A}^a T_{ij}^a\psi_j \end{aligned} \quad (286)$$

$$\mathcal{L} = i\bar{\psi}\not{\partial}\psi + g\bar{\psi}_i\mathbf{A}^a T_{ij}^a\psi_j - m\bar{\psi}\psi + \frac{(dA)^2}{2} - gf^{abc}A_\mu^b A_\nu^c \partial^\mu A^{a\nu} \quad (287)$$

$$\mathcal{L} = \bar{\psi}(i\not{\partial} - m)\psi + \frac{(dA)^2}{2} + g\bar{\psi}_i\mathbf{A}^a T_{ij}^a\psi_j - gf^{abc}A_\mu^b A_\nu^c \partial^\mu A^{a\nu} \quad (288)$$

$$\boxed{\mathcal{L}_0 = \bar{\psi}(i\not{\partial} - m)\psi + \frac{(dA)^2}{2}} \quad (289)$$

$$\boxed{\mathcal{L}_1 = g\bar{\psi}_i\mathbf{A}^a T_{ij}^a\psi_j - gf^{abc}A_\mu^b A_\nu^c \partial^\mu A^{a\nu}} \quad (290)$$

$$\boxed{\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_1} \quad (291)$$

18. FEYNMAN RULES

The quark (ψ) propagator is

$$j \longrightarrow \vec{p} \longrightarrow i \sim \frac{i\delta^{ij}}{\not{p} - m + i\epsilon}. \quad (292)$$

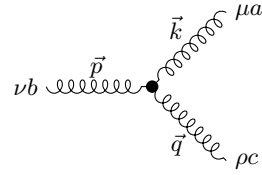


FIG. 17. The coefficient is $gf^{abc}[\eta^{\mu\nu}(\pm\vec{k} \mp \vec{p})^\rho + \eta^{\nu\rho}(\pm\vec{p} \mp \vec{q})^\mu + \eta^{\rho\mu}(\pm\vec{q} \mp \vec{k})^\nu]$, with the top/bottom sign for incoming/outgoing momenta. Note how the terms are generated by moving counterclockwise.

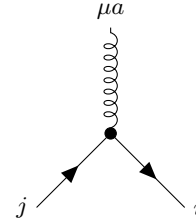


FIG. 18. The coefficient is $ig\gamma^\mu T_{ij}^a$, where the left/right indices i/j are assigned to outgoing/incoming lines.

The gluon (A) propagator is

$$\nu b \text{ wavy line } \vec{p} \text{ wavy line } \mu a \sim -\frac{i\eta^{\mu\nu}}{p^2 + i\epsilon}\delta^{ab}. \quad (293)$$

The two relevant interactions are in Figs. 17 & 18.

Real on-shell external quarks are capped with momentum eigenstate spinors (Table II). On-shell photons get polarization vectors (Schwartz Sec. 8.2.2). So do on shell gluons (Table III).

19. KINEMATICS

$$\vec{p}_1(q) + \vec{p}_2(\bar{q}) \rightarrow \vec{k}_1(g) + \vec{k}_2(g) \quad (294)$$

$$\vec{P} = \vec{p}_1 + \vec{p}_2 = \vec{k}_1 + \vec{k}_2 \quad (295)$$

$$s = P^2$$

$$t = (\vec{p}_1 - \vec{k}_1)^2 = (\vec{p}_2 - \vec{k}_2)^2 \quad (296)$$

$$u = (\vec{p}_1 - \vec{k}_2)^2 = (\vec{p}_2 - \vec{k}_1)^2$$

Particle	Spinor
incoming quark	u_s^i
outgoing quark	$\bar{u}_s^i$
incoming antiquark	$\bar{v}_s^i$
outgoing antiquark	v_s^i

TABLE II. Spinors for external quarks: $s = 1, 2$ is the spin index and i is the color index. The u and v spinors are defined as in QED (see my QED notes). Color is independent: $u_s^i = u_s\chi^i$.

Particle	Polarization Vector
incoming gluon	$\vec{\epsilon}_\sigma^a$
outgoing gluon	$\vec{\epsilon}_\sigma^{a*}$

TABLE III. Polarization vectors for external gluons. Same as photons in QED with color added independently: $\vec{\epsilon}_\sigma^a = \vec{\epsilon}_\sigma e^a$. There are 2 independent polarizations: $\sigma = 1, 2$.

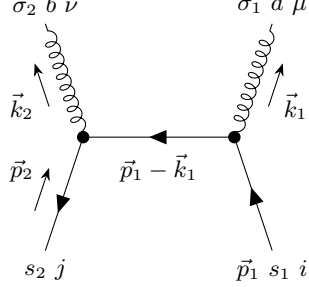


FIG. 19. ♠[t]

20. TREE LEVEL $\mathcal{M}$

There are 3 tree level diagrams (Figs. 19-21).

$$i\mathcal{M} = \spadesuit[s] + \spadesuit[t] + \spadesuit[u] \quad (297)$$

20.1. ♠[s]

$$\spadesuit[s] = \bar{v}_2^j i g \gamma^\alpha T_{ji}^d u_1^i \frac{-i\eta_{\alpha\beta}}{P^2 + i\epsilon} \delta^{cd} g f^{abc} [\eta^{\mu\nu}(-\vec{k}_1 + \vec{k}_2)^\beta + \eta^{\nu\beta}(-\vec{k}_2 - \vec{P})^\mu + \eta^{\beta\mu}(\vec{P} + \vec{k}_1)^\nu] \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} \quad (298)$$

$$\bar{v}_2^j = \bar{v}_{s_2}^j(\vec{p}_2), \quad u_1^i = u_{s_1}^i(\vec{p}_1), \quad \epsilon_{\mu 1}^a = \epsilon_{\mu\sigma_1}^a(\vec{k}_1), \quad \epsilon_{\nu 2}^b = \epsilon_{\nu\sigma_2}^b(\vec{k}_2) \quad (299)$$

$$\epsilon \rightarrow 0 \quad (300)$$

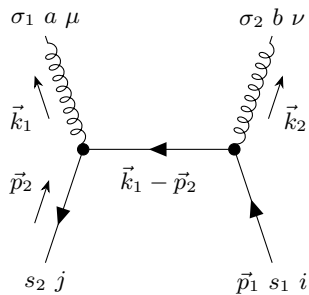


FIG. 20. ♠[u]

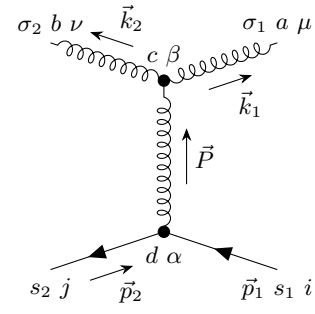


FIG. 21. ♠[s]

$$\spadesuit[s] = g^2 \bar{v}_2^j \gamma^\alpha T_{ji}^d u_1^i \frac{\eta_{\alpha\beta}}{s} \delta^{cd} f^{abc} \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} [\eta^{\mu\nu}(\vec{k}_2 - \vec{k}_1)^\beta - \eta^{\nu\beta}(\vec{k}_2 + \vec{P})^\mu + \eta^{\beta\mu}(\vec{P} + \vec{k}_1)^\nu] \quad (301)$$

I make sums over color indices explicit. We do not sum over external colors (i, j, a, b).

$$\spadesuit[s] = g^2 \sum_{cd=1}^8 \bar{v}_2^j \gamma^\alpha T_{ji}^d u_1^i \frac{\eta_{\alpha\beta}}{s} \delta^{cd} f^{abc} \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} [\eta^{\mu\nu}(\vec{k}_2 - \vec{k}_1)^\beta - \eta^{\nu\beta}(\vec{k}_2 + \vec{P})^\mu + \eta^{\beta\mu}(\vec{P} + \vec{k}_1)^\nu] \quad (302)$$

$$\spadesuit[s] = g^2 \sum_{c=1}^8 \bar{v}_2^j \gamma_\beta T_{ji}^c u_1^i \frac{1}{s} f^{abc} \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} [\eta^{\mu\nu}(\vec{k}_2 - \vec{k}_1)^\beta - \eta^{\nu\beta}(\vec{k}_2 + \vec{P})^\mu + \eta^{\beta\mu}(\vec{P} + \vec{k}_1)^\nu] \quad (303)$$

$$\spadesuit[s] = g^2 \sum_{c=1}^8 \bar{v}_2^j \gamma_\beta f^{abc} T_{ji}^c u_1^i \frac{1}{s} \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} [\eta^{\mu\nu}(\vec{k}_2 - \vec{k}_1)^\beta - \eta^{\nu\beta}(\vec{k}_2 + \vec{P})^\mu + \eta^{\beta\mu}(\vec{P} + \vec{k}_1)^\nu] \quad (304)$$

From Schwartz Eq. 25.8

$$[T^a, T^b] = i \sum_{c=1}^8 f^{abc} T^c. \quad (305)$$

$$\sum_{c=1}^8 f^{abc} T^c = i[T^b, T^a] \quad (306)$$

$$\spadesuit[s] = g^2 \frac{i}{s} \bar{v}_2^j \gamma_\beta [T^b, T^a]_{ji} u_1^i \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} [\eta^{\mu\nu}(\vec{k}_2 - \vec{k}_1)^\beta - \eta^{\nu\beta}(\vec{k}_2 + \vec{P})^\mu + \eta^{\beta\mu}(\vec{P} + \vec{k}_1)^\nu] \quad (307)$$

$$\spadesuit[s] = g^2 \frac{i}{s} [T^b, T^a]_{ji} \bar{v}_2^j \gamma_\beta u_1^i \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} [\eta^{\mu\nu}(\vec{k}_2 - \vec{k}_1)^\beta - \eta^{\nu\beta}(\vec{k}_2 + \vec{P})^\mu + \eta^{\beta\mu}(\vec{P} + \vec{k}_1)^\nu] \quad (308)$$

$$\spadesuit[s] = g^2 \frac{i}{s} [T^b, T^a]_{ji} \bar{v}_2^j \gamma_\beta [\eta^{\mu\nu} (\vec{k}_2 - \vec{k}_1)^\beta - \eta^{\nu\beta} (\vec{k}_2 + \vec{P})^\mu + \eta^{\beta\mu} (\vec{P} + \vec{k}_1)^\nu] u_1^i \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} \quad (309)$$

$$\spadesuit[s] = g^2 \frac{i}{s} [T^b, T^a]_{ji} \bar{v}_2^j [\eta^{\mu\nu} (\gamma_\beta k_2^\beta - \gamma_\beta k_1^\beta) - \gamma_\beta \eta^{\nu\beta} (\vec{k}_2 + \vec{P})^\mu + \gamma_\beta \eta^{\beta\mu} (\vec{P} + \vec{k}_1)^\nu] u_1^i \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} \quad (310)$$

$$\spadesuit[s] = g^2 \frac{i}{s} [T^b, T^a]_{ji} \bar{v}_2^j [\eta^{\mu\nu} (k_2 - k_1) - \gamma^\nu (\vec{k}_2 + \vec{P})^\mu + \gamma^\mu (\vec{P} + \vec{k}_1)^\nu] u_1^i \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} \quad (311)$$

$$\vec{P} = \vec{k}_1 + \vec{k}_2 \quad (312)$$

$$\spadesuit[s] = g^2 \frac{i}{s} [T^b, T^a]_{ji} \bar{v}_2^j [\eta^{\mu\nu} (k_2 - k_1) - \gamma^\nu (\vec{k}_1 + 2\vec{k}_2)^\mu + \gamma^\mu (2\vec{k}_1 + \vec{k}_2)^\nu] u_1^i \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} \quad (313)$$

$$\spadesuit[s] = g^2 \frac{i}{s} [T^b, T^a]_{ji} \bar{v}_2^j [\eta^{\mu\nu} (k_2 - k_1) - \gamma^\nu (k_1^\mu + 2k_2^\mu) + \gamma^\mu (2k_1^\nu + k_2^\nu)] u_1^i \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} \quad (314)$$

$$\spadesuit[s] = g^2 \frac{i}{s} [T^b, T^a]_{ji} \bar{v}_2^j [\eta^{\mu\nu} \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} (k_2 - k_1) - \gamma^\nu (k_1^\mu + 2k_2^\mu) \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} + \gamma^\mu (2k_1^\nu + k_2^\nu) \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*}] u_1^i \quad (315)$$

$$\spadesuit[s] = g^2 \frac{i}{s} [T^b, T^a]_{ji} \bar{v}_2^j [\bar{\epsilon}_1^{a*} \cdot \bar{\epsilon}_2^{b*} (k_2 - k_1) - \gamma^\nu \epsilon_{\nu 2}^{b*} (k_1^\mu \epsilon_{\mu 1}^{a*} + 2k_2^\mu \epsilon_{\mu 1}^{a*}) + \gamma^\mu \epsilon_{\mu 1}^{a*} (2k_1^\nu \epsilon_{\nu 2}^{b*} + k_2^\nu \epsilon_{\nu 2}^{b*})] u_1^i \quad (316)$$

$$\spadesuit[s] = g^2 \frac{i}{s} [T^b, T^a]_{ji} \bar{v}_2^j [\bar{\epsilon}_1^{a*} \cdot \bar{\epsilon}_2^{b*} (k_2 - k_1) - \not{\epsilon}_2^{b*} (\vec{k}_1 \cdot \bar{\epsilon}_1^{a*} + 2\vec{k}_2 \cdot \bar{\epsilon}_1^{a*}) + \not{\epsilon}_1^{a*} (2\vec{k}_1 \cdot \bar{\epsilon}_2^{b*} + \vec{k}_2 \cdot \bar{\epsilon}_2^{b*})] u_1^i \quad (317)$$

Polarization vectors are transverse: $\bar{\epsilon}(\vec{p}) \cdot \vec{p} = 0$.

$$\spadesuit[s] = i \frac{g^2}{s} [T^b, T^a]_{ji} \bar{v}_2^j [\bar{\epsilon}_1^{a*} \cdot \bar{\epsilon}_2^{b*} (k_2 - k_1) - 2\not{\epsilon}_2^{b*} \vec{k}_2 \cdot \bar{\epsilon}_1^{a*} + 2\not{\epsilon}_1^{a*} \vec{k}_1 \cdot \bar{\epsilon}_2^{b*}] u_1^i \quad (318)$$

20.2. ♠[t]

$$\spadesuit[t] = \sum_{nm=1}^3 \bar{v}_2^j i g \gamma^\nu T_{jn}^b \frac{i \delta^{nm}}{\not{p}_1 - \not{k}_1 - m + i\epsilon} i g \gamma^\mu T_{mi}^a u_1^i \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} \quad (319)$$

$$\spadesuit[t] = \frac{g^2}{i} \sum_{nm=1}^3 \bar{v}_2^j \gamma^\nu T_{jn}^b \frac{\delta^{nm}}{\not{p}_1 - \not{k}_1 - m} \gamma^\mu T_{mi}^a u_1^i \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} \quad (320)$$

$$\spadesuit[t] = \frac{g^2}{i} \sum_{n=1}^3 \bar{v}_2^j \gamma^\nu T_{jn}^b \frac{1}{\not{p}_1 - \not{k}_1 - m} \gamma^\mu T_{ni}^a u_1^i \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} \quad (321)$$

$$\spadesuit[t] = \frac{g^2}{i} \sum_{n=1}^3 T_{jn}^b T_{ni}^a \bar{v}_2^j \gamma^\nu \frac{1}{\not{p}_1 - \not{k}_1 - m} \gamma^\mu u_1^i \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} \quad (322)$$

$$\sum_{n=1}^3 T_{jn}^b T_{ni}^a = (T^b T^a)_{ji} \quad (323)$$

$$\spadesuit[t] = \frac{g^2}{i} (T^b T^a)_{ji} \bar{v}_2^j \gamma^\nu \frac{1}{\not{p}_1 - \not{k}_1 - m} \gamma^\mu u_1^i \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} \quad (324)$$

$$\frac{1}{\not{p} - m} = \frac{\not{p} + m}{(\not{p} - m)(\not{p} + m)} = \frac{\not{p} + m}{\not{p}^2 - m^2} \quad (325)$$

$$\not{p}^2 = \gamma^\mu p_\mu \gamma^\nu p_\nu = \gamma^\mu \gamma^\nu p_\mu p_\nu \quad (326)$$

$$2\gamma^\mu \gamma^\nu = \gamma^\mu \gamma^\nu + \gamma^\mu \gamma^\nu = \gamma^\mu \gamma^\nu - \gamma^\nu \gamma^\mu + \gamma^\nu \gamma^\mu + \gamma^\mu \gamma^\nu = [\gamma^\mu, \gamma^\nu] + \{\gamma^\mu, \gamma^\nu\} \quad (327)$$

$$\gamma^\mu \gamma^\nu = \frac{[\gamma^\mu, \gamma^\nu] + \{\gamma^\mu, \gamma^\nu\}}{2} \quad (328)$$

$$\not{p}^2 = \frac{[\gamma^\mu, \gamma^\nu] p_\mu p_\nu + \{\gamma^\mu, \gamma^\nu\} p_\mu p_\nu}{2} \quad (329)$$

$$[\gamma^\mu, \gamma^\nu] p_\mu p_\nu = \not{p}^2 - \not{p}^2 = 0 \quad (330)$$

$$\not{p}^2 = \frac{\{\gamma^\mu, \gamma^\nu\} p_\mu p_\nu}{2} \quad (331)$$

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} \quad (332)$$

$$\not{p}^2 = \eta^{\mu\nu} p_\mu p_\nu = \vec{p} \cdot \vec{p} = p^2 \quad (333)$$

$$\frac{1}{\not{p} - m} = \frac{\not{p} + m}{p^2 - m^2} \quad (334)$$

$$\spadesuit[t] = \frac{g^2}{i} (T^b T^a)_{ji} \bar{v}_2^j \gamma^\nu \frac{\not{p}_1 - \not{k}_1 + m}{(\vec{p}_1 - \vec{k}_1)^2 - m^2} \gamma^\mu u_1^i \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} \quad (335)$$

$$t = (\vec{p}_1 - \vec{k}_1)^2 \quad (336)$$

$$\spadesuit[t] = \frac{g^2}{i} (T^b T^a)_{ji} \bar{v}_2^j \gamma^\nu \frac{\not{p}_1 - \not{k}_1 + m}{t - m^2} \gamma^\mu u_1^i \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} \quad (337)$$

$$\spadesuit[t] = \frac{g^2 (T^b T^a)_{ji}}{i(t - m^2)} \bar{v}_2^j \gamma^\nu (\not{p}_1 \gamma^\mu u_1^i - \not{k}_1 \gamma^\mu u_1^i + m \gamma^\mu u_1^i) \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} \quad (338)$$

$$\spadesuit[t] = \frac{g^2 (T^b T^a)_{ji}}{i(t - m^2)} \bar{v}_2^j \gamma^\nu (\not{p}_1 \gamma^\mu u_1^i - \not{k}_1 \gamma^\mu u_1^i + m \gamma^\mu u_1^i) \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} \quad (339)$$

$$\not{p} \gamma^\mu = p_\alpha \gamma^\alpha \gamma^\mu = p_\alpha (2\eta^{\alpha\mu} - \gamma^\mu \gamma^\alpha) = 2\eta^{\alpha\mu} p_\alpha - \gamma^\mu \gamma^\alpha p_\alpha = 2p^\mu - \gamma^\mu \not{p} \quad (340)$$

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} \implies \gamma^\mu \gamma^\nu = 2\eta^{\mu\nu} - \gamma^\nu \gamma^\mu \quad (341)$$

$$\spadesuit[t] = \frac{g^2 (T^b T^a)_{ji}}{i(t - m^2)} \bar{v}_2^j \gamma^\nu (2p_1^\mu u_1^i - \gamma^\mu \not{p}_1 u_1^i - \not{k}_1 \gamma^\mu u_1^i + m \gamma^\mu u_1^i) \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} \quad (342)$$

$$\not{p} u(\vec{p}) = m u(\vec{p}) \quad (343)$$

$$\spadesuit[t] = \frac{g^2 (T^b T^a)_{ji}}{i(t - m^2)} \bar{v}_2^j \gamma^\nu (2p_1^\mu u_1^i - m \gamma^\mu u_1^i - \not{k}_1 \gamma^\mu u_1^i + m \gamma^\mu u_1^i) \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} \quad (344)$$

$$\spadesuit[t] = \frac{g^2 (T^b T^a)_{ji}}{i(t - m^2)} \bar{v}_2^j \gamma^\nu (2p_1^\mu u_1^i - \not{k}_1 \gamma^\mu u_1^i) \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} \quad (345)$$

$$\spadesuit[t] = \frac{g^2 (T^b T^a)_{ji}}{i(t - m^2)} \bar{v}_2^j \gamma^\nu (2p_1^\mu - \not{k}_1 \gamma^\mu) u_1^i \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} \quad (346)$$

$$\spadesuit[t] = \frac{g^2 (T^b T^a)_{ji}}{i(t - m^2)} \bar{v}_2^j \gamma^\nu \epsilon_{\nu 2}^{b*} (2p_1^\mu \epsilon_{\mu 1}^{a*} - \not{k}_1 \gamma^\mu \epsilon_{\mu 1}^{a*}) u_1^i \quad (347)$$

$$\spadesuit[t] = \frac{ig^2 (T^b T^a)_{ji}}{m^2 - t} \bar{v}_2^j \epsilon_2^{b*} (2\vec{p}_1 \cdot \vec{\epsilon}_1^{a*} - \not{k}_1 \not{\epsilon}_1^{a*}) u_1^i \quad (348)$$

20.3. ♠[u]

$$\spadesuit[u] = \sum_{nm=1}^3 \bar{v}_2^j ig \gamma^\mu T_{jn}^a \frac{i \delta^{nm}}{\not{k}_1 - \not{p}_2 - m + i\epsilon} ig \gamma^\nu T_{mi}^b u_1^i \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} \quad (349)$$

$$\spadesuit[u] = \frac{g^2}{i} \bar{v}_2^j \gamma^\mu \frac{\sum_{nm=1}^3 T_{jn}^a T_{mi}^b \delta^{nm}}{\not{k}_1 - \not{p}_2 - m} \gamma^\nu u_1^i \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} \quad (350)$$

$$\sum_{nm=1}^3 T_{jn}^a T_{mi}^b \delta^{nm} = \sum_{n=1}^3 T_{jn}^a T_{ni}^b = (T^a T^b)_{ji} \quad (351)$$

$$\spadesuit[u] = \frac{g^2}{i} (T^a T^b)_{ji} \bar{v}_2^j \gamma^\mu \frac{1}{\not{k}_1 - \not{p}_2 - m} \gamma^\nu u_1^i \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} \quad (352)$$

$$\spadesuit[u] = \frac{g^2}{i} (T^a T^b)_{ji} \bar{v}_2^j \gamma^\mu \frac{\not{k}_1 - \not{p}_2 + m}{(\vec{k}_1 - \vec{p}_2)^2 - m^2} \gamma^\nu u_1^i \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} \quad (353)$$

$$u = (\vec{p}_2 - \vec{k}_1)^2 = (\vec{k}_1 - \vec{p}_2)^2 \quad (354)$$

$$\spadesuit[u] = ig^2 (T^a T^b)_{ji} \bar{v}_2^j \gamma^\mu \frac{\not{k}_1 - \not{p}_2 + m}{m^2 - u} \gamma^\nu u_1^i \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} \quad (355)$$

$$\spadesuit[u] = \frac{ig^2 (T^a T^b)_{ji}}{m^2 - u} \bar{v}_2^j \gamma^\mu (\not{k}_1 - \not{p}_2 + m) \gamma^\nu u_1^i \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} \quad (356)$$

$$\spadesuit[u] = \frac{ig^2 (T^a T^b)_{ji}}{m^2 - u} \bar{v}_2^j (\gamma^\mu \not{k}_1 - \gamma^\mu \not{p}_2 + m \gamma^\mu) \gamma^\nu u_1^i \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} \quad (357)$$

$$\not{p} \gamma^\mu = 2p^\mu - \gamma^\mu \not{p} \implies \gamma^\mu \not{p} = 2p^\mu - \not{p} \gamma^\mu \quad (358)$$

$$\spadesuit[u] = \frac{ig^2 (T^a T^b)_{ji}}{m^2 - u} \bar{v}_2^j (\gamma^\mu \not{k}_1 - 2p_2^\mu + \not{p}_2 \gamma^\mu + m \gamma^\mu) \gamma^\nu u_1^i \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} \quad (359)$$

$$\spadesuit[u] = \frac{ig^2 (T^a T^b)_{ji}}{m^2 - u} \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} (\bar{v}_2^j \gamma^\mu \not{k}_1 - 2p_2^\mu \bar{v}_2^j + \bar{v}_2^j \not{p}_2 \gamma^\mu + m \bar{v}_2^j \gamma^\mu) \gamma^\nu u_1^i \quad (360)$$

$$\bar{v}(\vec{p})(\not{p} + m) = 0 \implies \bar{v} \not{p} = -m \bar{v} \quad (361)$$

$$\spadesuit[u] = \frac{ig^2 (T^a T^b)_{ji}}{m^2 - u} \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} (\bar{v}_2^j \gamma^\mu \not{k}_1 - 2p_2^\mu \bar{v}_2^j - m \bar{v}_2^j \gamma^\mu + m \bar{v}_2^j \gamma^\mu) \gamma^\nu u_1^i \quad (362)$$

$$\spadesuit[u] = \frac{ig^2 (T^a T^b)_{ji}}{m^2 - u} \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} (\bar{v}_2^j \gamma^\mu \not{k}_1 - 2p_2^\mu \bar{v}_2^j) \gamma^\nu u_1^i \quad (363)$$

$$\spadesuit[u] = \frac{ig^2 (T^a T^b)_{ji}}{m^2 - u} \bar{v}_2^j \epsilon_{\mu 1}^{a*} \epsilon_{\nu 2}^{b*} (\gamma^\mu \not{k}_1 - 2p_2^\mu) \gamma^\nu u_1^i \quad (364)$$

$$\spadesuit[u] = \frac{ig^2 (T^a T^b)_{ji}}{m^2 - u} \bar{v}_2^j (\not{\epsilon}_1^{a*} \not{k}_1 - 2\vec{p}_2 \cdot \vec{\epsilon}_1^{a*}) \not{\epsilon}_2^{b*} u_1^i \quad (365)$$

20.4. $\mathcal{M}$

$$i\mathcal{M} = \spadesuit[s] + \spadesuit[t] + \spadesuit[u] \quad (366)$$

$$\begin{aligned} \frac{\mathcal{M}}{g^2} &= \frac{(T^a T^b)_{ji}}{m^2 - u} \bar{v}_2^j (\not{\epsilon}_1^{a*} \not{k}_1 - 2\vec{p}_2 \cdot \vec{\epsilon}_1^{a*}) \not{\epsilon}_2^{b*} u_1^i \\ &+ \frac{(T^b T^a)_{ji}}{m^2 - t} \bar{v}_2^j \not{\epsilon}_2^{b*} (2\vec{p}_1 \cdot \vec{\epsilon}_1^{a*} - \not{k}_1 \not{\epsilon}_1^{a*}) u_1^i + \frac{[T^b, T^a]_{ji}}{s} \bar{v}_2^j \\ &[\vec{\epsilon}_1^{a*} \cdot \vec{\epsilon}_2^{b*} (\not{k}_2 - \not{k}_1) - 2\not{\epsilon}_2^{b*} \vec{k}_2 \cdot \vec{\epsilon}_1^{a*} + 2\not{\epsilon}_1^{a*} \vec{k}_1 \cdot \vec{\epsilon}_2^{b*}] u_1^i \end{aligned} \quad (367)$$

21. WARD IDENTITY CHECK

I replace external gluon 1's polarization vector with its momentum

$$\heartsuit = M(\vec{\epsilon}_1^{a*} \rightarrow \vec{k}_1). \quad (368)$$

The Ward identity says $\heartsuit = 0$.

$$\begin{aligned} \frac{\mathcal{M}}{g^2} &= \frac{(T^a T^b)_{ji}}{m^2 - u} \bar{v}_2^j (\not{k}_1^2 - 2\vec{p}_2 \cdot \vec{k}_1) \not{\epsilon}_2^{b*} u_1^i \\ &+ \frac{(T^b T^a)_{ji}}{m^2 - t} \bar{v}_2^j \not{\epsilon}_2^{b*} (2\vec{p}_1 \cdot \vec{k}_1 - \not{k}_1^2) u_1^i + \frac{[T^b, T^a]_{ji}}{s} \bar{v}_2^j \\ &[\vec{k}_1 \cdot \vec{\epsilon}_2^{b*} (\not{k}_2 - \not{k}_1) - 2\not{\epsilon}_2^{b*} \vec{k}_2 \cdot \vec{k}_1 + 2\not{k}_1 \vec{k}_1 \cdot \vec{\epsilon}_2^{b*}] u_1^i \end{aligned} \quad (369)$$

$$\begin{aligned} \heartsuit &\propto \frac{(T^a T^b)_{ji}}{m^2 - u} \bar{v}_2^j (\not{k}_1^2 - 2\vec{p}_2 \cdot \vec{k}_1) \not{\epsilon}_2^{b*} u_1^i \\ &+ \frac{(T^b T^a)_{ji}}{m^2 - t} \bar{v}_2^j \not{\epsilon}_2^{b*} (2\vec{p}_1 \cdot \vec{k}_1 - \not{k}_1^2) u_1^i + \frac{[T^b, T^a]_{ji}}{s} \bar{v}_2^j \\ &[\vec{k}_1 \cdot \vec{\epsilon}_2^{b*} (\not{k}_2 - \not{k}_1) - 2\not{\epsilon}_2^{b*} \vec{k}_2 \cdot \vec{k}_1 + 2\not{k}_1 \vec{k}_1 \cdot \vec{\epsilon}_2^{b*}] u_1^i \end{aligned} \quad (370)$$

$$\not{k}_1^2 = k_1^2 = 0 \quad (371)$$

$$\begin{aligned} \heartsuit &\propto 2 \left[\frac{(T^b T^a)_{ji}}{m^2 - t} \vec{p}_1 \cdot \vec{k}_1 - \frac{(T^a T^b)_{ji}}{m^2 - u} \vec{p}_2 \cdot \vec{k}_1 \right] \bar{v}_2^j \not{\epsilon}_2^{b*} u_1^i \\ &+ \frac{[T^b, T^a]_{ji}}{s} \bar{v}_2^j [\vec{k}_1 \cdot \vec{\epsilon}_2^{b*} (\not{k}_2 - \not{k}_1) - 2\not{\epsilon}_2^{b*} \vec{k}_2 \cdot \vec{k}_1 + 2\not{k}_1 \vec{k}_1 \cdot \vec{\epsilon}_2^{b*}] u_1^i \end{aligned} \quad (372)$$

$$\begin{aligned} \heartsuit &\propto 2 \left[\frac{(T^b T^a)_{ji}}{m^2 - t} \vec{p}_1 \cdot \vec{k}_1 - \frac{(T^a T^b)_{ji}}{m^2 - u} \vec{p}_2 \cdot \vec{k}_1 \right] \bar{v}_2^j \not{\epsilon}_2^{b*} u_1^i \\ &+ \frac{[T^b, T^a]_{ji}}{s} \bar{v}_2^j [(\not{k}_2 - \not{k}_1 + 2\not{k}_1) \vec{k}_1 \cdot \vec{\epsilon}_2^{b*} - 2\not{\epsilon}_2^{b*} \vec{k}_2 \cdot \vec{k}_1] u_1^i \end{aligned} \quad (373)$$

$$\begin{aligned} \heartsuit &\propto 2 \left[\frac{(T^b T^a)_{ji}}{m^2 - t} \vec{p}_1 \cdot \vec{k}_1 - \frac{(T^a T^b)_{ji}}{m^2 - u} \vec{p}_2 \cdot \vec{k}_1 \right] \bar{v}_2^j \not{\epsilon}_2^{b*} u_1^i \\ &+ \frac{[T^b, T^a]_{ji}}{s} \bar{v}_2^j [(\not{k}_2 + \not{k}_1) \vec{k}_1 \cdot \vec{\epsilon}_2^{b*} - 2\not{\epsilon}_2^{b*} \vec{k}_2 \cdot \vec{k}_1] u_1^i \end{aligned} \quad (374)$$

$$\vec{P} = \vec{k}_1 + \vec{k}_2 \quad (375)$$

$$\begin{aligned} \heartsuit &\propto 2 \left[\frac{(T^b T^a)_{ji}}{m^2 - t} \vec{p}_1 \cdot \vec{k}_1 - \frac{(T^a T^b)_{ji}}{m^2 - u} \vec{p}_2 \cdot \vec{k}_1 \right] \bar{v}_2^j \not{\epsilon}_2^{b*} u_1^i \\ &+ \frac{[T^b, T^a]_{ji}}{s} \bar{v}_2^j (\not{P} \vec{k}_1 \cdot \vec{\epsilon}_2^{b*} - 2\not{\epsilon}_2^{b*} \vec{k}_2 \cdot \vec{k}_1) u_1^i \end{aligned} \quad (376)$$

$$s = P^2 = (\vec{k}_1 + \vec{k}_2)^2 = k_1^2 + k_2^2 + 2\vec{k}_1 \cdot \vec{k}_2 \quad (377)$$

$$k_1^2 = k_2^2 = 0, \quad p_1^2 = p_2^2 = m^2 \quad (378)$$

$$s = 2\vec{k}_1 \cdot \vec{k}_2 \quad (379)$$

$$\begin{aligned} \heartsuit &\propto 2 \left[\frac{(T^b T^a)_{ji}}{m^2 - t} \vec{p}_1 \cdot \vec{k}_1 - \frac{(T^a T^b)_{ji}}{m^2 - u} \vec{p}_2 \cdot \vec{k}_1 \right] \bar{v}_2^j \not{\epsilon}_2^{b*} u_1^i \\ &+ \frac{[T^b, T^a]_{ji}}{s} \bar{v}_2^j (\not{P} \vec{k}_1 \cdot \vec{\epsilon}_2^{b*} - s \not{\epsilon}_2^{b*}) u_1^i \end{aligned} \quad (380)$$

$$t = (\vec{p}_1 - \vec{k}_1)^2 = p_1^2 + k_1^2 - 2\vec{p}_1 \cdot \vec{k}_1 = m^2 - 2\vec{p}_1 \cdot \vec{k}_1 \quad (381)$$

$$2\vec{p}_1 \cdot \vec{k}_1 = m^2 - t \quad (382)$$

$$u = (\vec{p}_2 - \vec{k}_1)^2 = m^2 - 2\vec{p}_2 \cdot \vec{k}_1 \implies 2\vec{p}_2 \cdot \vec{k}_1 = m^2 - u \quad (383)$$

$$\begin{aligned} \heartsuit &\propto [(T^b T^a)_{ji} - (T^a T^b)_{ji}] \bar{v}_2^j \not{\epsilon}_2^{b*} u_1^i \\ &+ \frac{[T^b, T^a]_{ji}}{s} \bar{v}_2^j (\not{P} \vec{k}_1 \cdot \vec{\epsilon}_2^{b*} - s \not{\epsilon}_2^{b*}) u_1^i \end{aligned} \quad (384)$$

$$\heartsuit \propto [T^b, T^a]_{ji} \bar{v}_2^j \not{\epsilon}_2^{b*} u_1^i + \frac{[T^b, T^a]_{ji}}{s} \bar{v}_2^j (\not{P} \vec{k}_1 \cdot \vec{\epsilon}_2^{b*} - s \not{\epsilon}_2^{b*}) u_1^i \quad (385)$$

$$\heartsuit \propto \bar{v}_2^j \not{\epsilon}_2^{b*} u_1^i + \frac{\bar{v}_2^j (\not{P} \vec{k}_1 \cdot \vec{\epsilon}_2^{b*} - s \not{\epsilon}_2^{b*}) u_1^i}{s} \quad (386)$$

$$\heartsuit \propto \bar{v}_2^j \not{\epsilon}_2^{b*} u_1^i + \frac{\vec{k}_1 \cdot \vec{\epsilon}_2^{b*}}{s} \bar{v}_2^j \not{P} u_1^i - \frac{s}{s} \bar{v}_2^j \not{\epsilon}_2^{b*} u_1^i \quad (387)$$

$$\heartsuit \propto \bar{v}_2^j \not{\epsilon}_2^{b*} u_1^i - \bar{v}_2^j \not{\epsilon}_2^{b*} u_1^i + \frac{\vec{k}_1 \cdot \vec{\epsilon}_2^{b*}}{s} \bar{v}_2^j \not{P} u_1^i \quad (388)$$

$$\heartsuit \propto \frac{\vec{k}_1 \cdot \vec{\epsilon}_2^{b*}}{s} \bar{v}_2^j \not{P} u_1^i \propto \bar{v}_2^j \not{P} u_1^i = \bar{v}_2^j (\not{p}_1 + \not{p}_2) u_1^i \quad (389)$$

$$\heartsuit \propto \bar{v}_2^j \not{p}_1 u_1^i + \bar{v}_2^j \not{p}_2 u_1^i \quad (390)$$

$$\not{p} u = m u, \quad \bar{v} \not{p} = -m \bar{v} \quad (391)$$

$$\heartsuit \propto m \bar{v}_2^j u_1^i - m \bar{v}_2^j u_1^i = 0 \implies \heartsuit = 0 \quad (392)$$

$\mathcal{M}$ satisfies the Ward identity. QED.